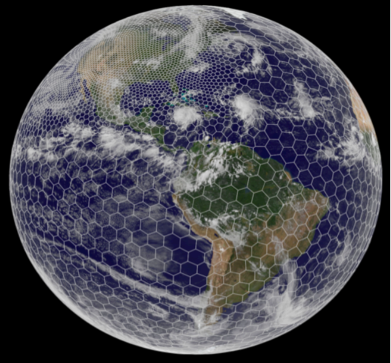




[Home](#)

/ [MPAS Documentation and Resources](#)

/ [MPAS Atmosphere User Guide](#)

/ Appendix D: Description of Model Fields



Appendix D: Description of Model Fields

Every field that may be read or written in a NetCDF *stream* (as described in [Configuring Model Input and Output](#)) by the MPAS-Atmosphere model is described in this chapter. The dimensionality of each field is given in Fortran storage order (i.e., the fastest-varying dimension is inner-most).

a_tri (real) (nVertLevels, nCells, Time)

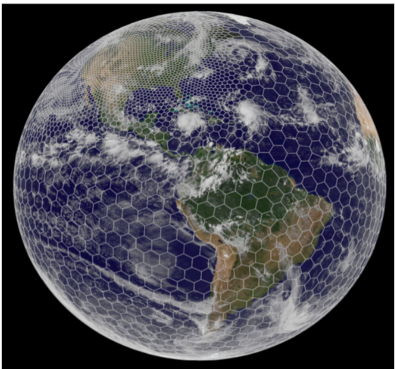
Units	
Description	implicit tridiagonal solve coefficients
Accessed in code	as a_tri from the diag

absnxt (real) (nVertLevels, cam_dim1, nCells, Time)

Units	
Description	Total nearest layer absorptivity
Accessed in code	as absnxt from the diag_physics

abstot (real) (nVertLevelsP1, nVertLevelsP1, nCells, Time)

Units	
-------	--



[MPAS Atmosphere](#)

Menu

Description	Total absorptivity
Accessed in code	as abstot from the diag_physics

aclwdnb (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky downward surface longwave radiation flux
Accessed in code	as aclwdnb from the diag_physics

aclwdnbc (real) (nCells, Time)

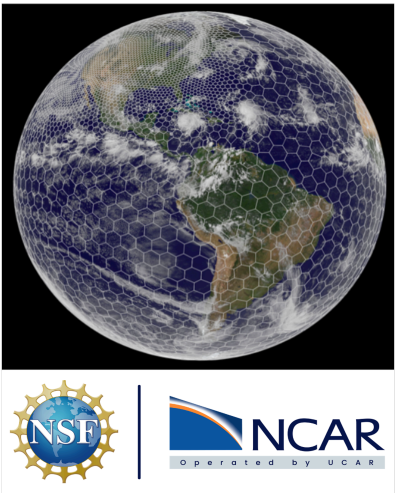
Units	W m ⁻²
Description	accumulated clear-sky downward surface longwave radiation flux
Accessed in code	as aclwdnbc from the diag_physics

aclwdnt (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky downward top-of-the-atmosphere longwave radiation flux
Accessed in code	as aclwdnt from the diag_physics

aclwdntc (real) (nCells, Time)

Units	W m ⁻²
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	accumulated clear-sky downward top-of-the-atmosphere longwave radiation flux
Accessed in code	as aclwdntc from the diag_physics

aclwupb (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky upward surface longwave radiation flux
Accessed in code	as aclwupb from the diag_physics

aclwupbc (real) (nCells, Time)

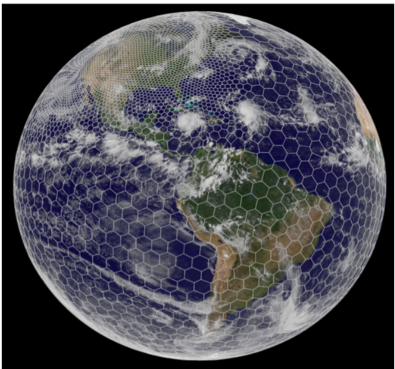
Units	W m ⁻²
Description	accumulated clear-sky upward surface longwave radiation flux
Accessed in code	as aclwupbc from the diag_physics

aclwupt (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky upward top-of-the-atmosphere longwave radiation flux
Accessed in code	as aclwupt from the diag_physics

aclwuptc (real) (nCells, Time)

Units	W m ⁻²
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	accumulated clear-sky upward top- of-the- atmosphere longwave radiation flux
Accessed in code	as aclwuptc from the diag_physics

acsnom (real) (nCells, Time)

Units	kg m ⁻²
Description	accumulated melted snow
Accessed in code	as acsnom from the diag_physics

acsnow (real) (nCells, Time)

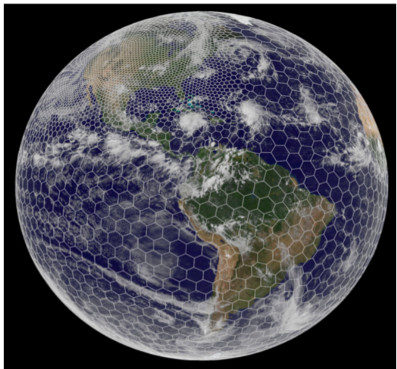
Units	kg m ⁻²
Description	accumulated snow
Accessed in code	as acsnow from the diag_physics

acswdnb (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky downward surface shortwave radiation flux
Accessed in code	as acswdnb from the diag_physics

acswdnbc (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated clear-sky downward surface shortwave radiation flux
Accessed in code	as acswdnbc from the diag_physics



[MPAS Atmosphere](#)

Menu

acswdnt (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky downward top-of-atmosphere shortwave radiation flux
Accessed in code	as acswdnt from the diag_physics

acswdntc (real) (nCells, Time)

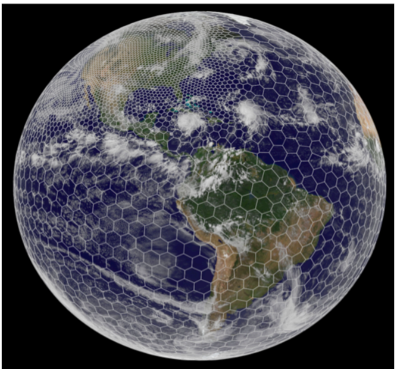
Units	W m ⁻²
Description	accumulated clear-sky downward top-of-atmosphere shortwave radiation flux
Accessed in code	as acswdntc from the diag_physics

acswupb (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky upward surface shortwave radiation flux
Accessed in code	as acswupb from the diag_physics

acswupbc (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated clear-sky upward surface shortwave radiation flux
Accessed in code	as acswupbc from the diag_physics



[MPAS Atmosphere](#)

Menu

acswupt (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated all-sky upward top- of- atmosphere shortwave radiation flux
Accessed in code	as acswupt from the diag_physics

acswuptc (real) (nCells, Time)

Units	W m ⁻²
Description	accumulated clear-sky upward top- of- atmosphere shortwave radiation flux
Accessed in code	as acswuptc from the diag_physics

adv_coefs (real) (FIFTEEN, nEdges)

Units	unitless
Description	Weighting coefficients used for reconstructing cell-based fields at edges
Accessed in code	as adv_coefs from the mesh

adv_coefs_3rd (real) (FIFTEEN, nEdges)

Units	unitless
Description	Weighting coefficients used for reconstructing cell-based fields at edges
Accessed in code	as adv_coefs_3rd from the mesh

advCellsForEdge (integer) (FIFTEEN, nEdges)

Units	•
-------	---



[MPAS Atmosphere](#)

Menu

Description	Cells used to reconstruct a cell-based field at an edge
Accessed in code	as <code>advCellsForEdge</code> from the mesh

aerosols (real) (nAerLevels, nCells, Time)

Description	aerosols is a variable array used in code which is made up of the constituent fields listed below.
Accessed in code	aerosols from the state pool

sul – constituent field of var_array aerosols

Description	Sulfate soluble (SUL) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

ssl – constituent field of var_array aerosols

Description	Sea-salt (SSLT) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

dust1 – constituent field of var_array aerosols

Description	Dust type 1 (DUST1) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

dust2 – constituent field of var_array aerosols



[MPAS Atmosphere](#)

Menu

Description	Dust type 2 (DUST2) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

dust3 – constituent field of var_array aerosols

Description	Dust type 3 (DUST3) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

dust4 – constituent field of var_array aerosols

Description	Dust type 4 (DUST4) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

ocpho – constituent field of var_array aerosols

Description	Hydrophobic organic carbon (OCPHO) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

bcpho – constituent field of var_array aerosols

Description	Hydrophobic black carbon (BCPHO) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

ocphi – constituent field of var_array aerosols



[MPAS Atmosphere](#)

Menu

Description	Hydrophilic organic carbon (OCPHI) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

bcphi – constituent field of var_array aerosols

Description	Hydrophilic black carbon (BCPHI) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

bg – constituent field of var_array aerosols

Description	Background (BG) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

volc – constituent field of var_array aerosols

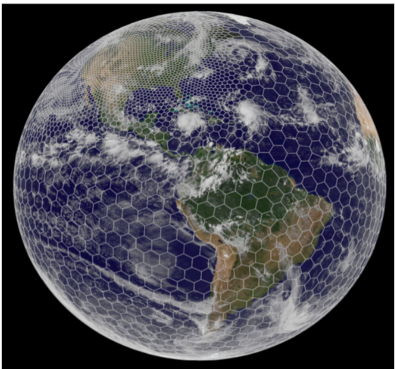
Description	Volcanic (VOLC) aerosol concentration
Units	kg m ⁻³
Array Group	aer_cam

albedo12m (real) (nMonths, nCells)

Units	unitless
Description	monthly-mean climatological surface albedo
Accessed in code	as albedo12m from the sfc_input

alpha_tri (real) (nVertLevels, nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	implicit tridiagonal solve coefficients
Accessed in code	as alpha_tri from the diag

angleEdge (real) (nEdges)

Units	rad
Description	Angle between local north and the positive tangential direction of an edge
Accessed in code	as angleEdge from the mesh

areaCell (real) (nCells)

Units	m ²
Description	Spherical area of a Voronoi cell
Accessed in code	as areaCell from the mesh

areaTriangle (real) (nVertices)

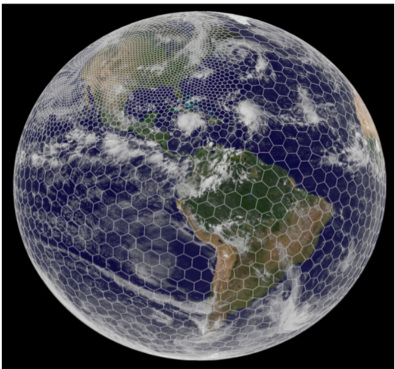
Units	m ²
Description	Spherical area of a Delaunay triangle
Accessed in code	as areaTriangle from the mesh

bdyMaskCell (integer) (nCells)

Units	•
Description	Limited-area specified/relaxation zone index for cells
Accessed in code	as bdyMaskCell from the mesh

bdyMaskEdge (integer) (nEdges)

Units	•
-------	---



[MPAS Atmosphere](#)

Menu

Description	Limited-area specified/relaxation zone index for edges
Accessed in code	as bdyMaskEdge from the mesh

bdyMaskVertex (integer) (nVertices)

Units	•
Description	Limited-area specified/relaxation zone index for vertices
Accessed in code	as bdyMaskVertex from the mesh

br (real) (nCells, Time)

Units	unitless
Description	Richardson number
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as br from the diag_physics

canwat (real) (nCells, Time)

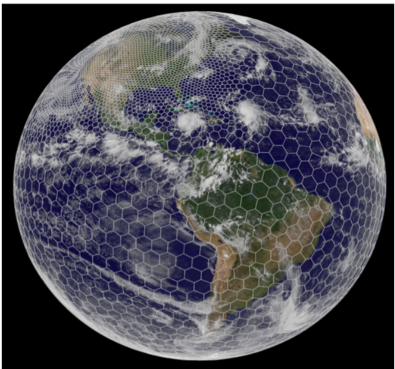
Units	kg m ⁻²
Description	water in canopy
Accessed in code	as canwat from the diag_physics

cape (real) (nCells, Time)

Units	J kg ⁻¹
Description	Convective available potential energy
Accessed in code	as cape from the diag

cd (real) (nCells, Time)

Units	unitless
Description	drag coefficient at 10-meter
Allocated by	bl_mynn_in, bl_ysu_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as cd from the diag_physics
------------------	-----------------------------

cda (real) (nCells, Time)

Units	unitless
Description	drag coefficient at lowest model level
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as cda from the diag_physics

cellsOnCell (integer) (maxEdges, nCells)

Units	•
Description	IDs of cells neighboring a cell
Accessed in code	as cellsOnCell from the mesh

cellsOnEdge (integer) (TWO, nEdges)

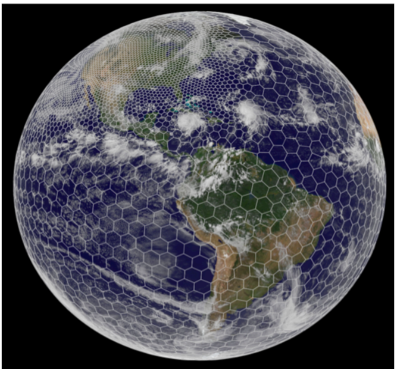
Units	•
Description	IDs of cells divided by an edge
Accessed in code	as cellsOnEdge from the mesh

cellsOnVertex (integer) (vertexDegree, nVertices)

Units	•
Description	IDs of the cells that meet at a vertex
Accessed in code	as cellsOnVertex from the mesh

cellTangentPlane (real) (R3, TWO, nCells)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	Components of a pair of vectors defining the tangent plane at a cell
Accessed in code	as cellTangentPlane from the mesh

cf1 (real) ()

Units	unitless
Description	Surface interpolation weight for level k=1 value
Accessed in code	as cf1 from the mesh

cf2 (real) ()

Units	unitless
Description	Surface interpolation weight for level k=2 value
Accessed in code	as cf2 from the mesh

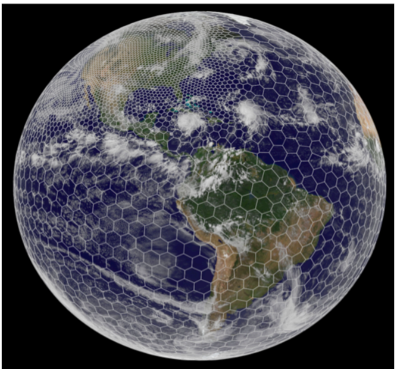
cf3 (real) ()

Units	unitless
Description	Surface interpolation weight for level k=3 value
Accessed in code	as cf3 from the mesh

ch (real) (nCells, Time)

Units	m s ⁻¹
Description	surface exchange coefficient for heat
Allocated by	bl_mynn_in
Accessed in code	as ch from the diag_physics

chklowq (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	unitless
Description	surface saturation flag
Accessed in code	as chklowq from the diag_physics

chs (real) (nCells, Time)

Units	m s^{-1}
Description	surface exchange coefficient for heat and moisture
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as chs from the diag_physics

chs2 (real) (nCells, Time)

Units	m s^{-1}
Description	surface exchange coefficient for heat at 2-meter
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as chs2 from the diag_physics

cin (real) (nCells, Time)

Units	J kg^{-1}
Description	Convective inhibition
Accessed in code	as cin from the diag

circulation (real) (nVertLevels, nVertices, Time)

Units	$\text{m}^2 \text{s}^{-1}$
Description	Horizontal circulation at vertices
Accessed in code	as circulation from the diag

ck (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	unitless
Description	enthalpy exchange coeff at 10-meter
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as ck from the diag_physics

cka (real) (nCells, Time)

Units	unitless
Description	enthalpy exchange coefficient at lowest model level
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as cka from the diag_physics

cldfrac (real) (nVertLevels, nCells, Time)

Units	unitless
Description	horizontal cloud fraction
Accessed in code	as cldfrac from the diag_physics

cldfrac_bl (real) (nVertLevels, nCells, Time)

Units	unitless
Description	cloud fraction in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as cldfrac_bl from the diag_physics

cldfrac_high_UPP (real) (nCells, Time)

Units	unitless
Description	Cloud fraction at high levels using UPP method
Accessed in code	as cldfrac_high_UPP from the diag



[MPAS](#)
[Atmosphere](#)

Menu

cldfrac_low_UPP (real) (nCells, Time)

Units	unitless
Description	Cloud fraction at low levels using UPP method
Accessed in code	as cldfrac_low_UPP from the diag

cldfrac_mid_UPP (real) (nCells, Time)

Units	unitless
Description	Cloud fraction at mid levels using UPP method
Accessed in code	as cldfrac_mid_UPP from the diag

cldfrac_tot_UPP (real) (nCells, Time)

Units	unitless
Description	Total cloud fraction using UPP column max method
Accessed in code	as cldfrac_tot_UPP from the diag

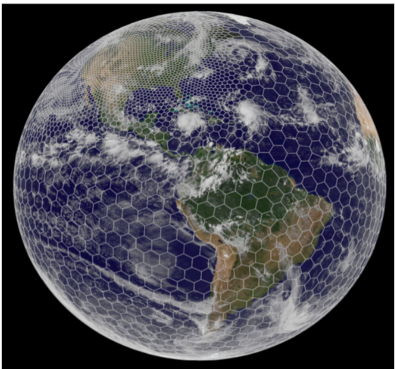
coeffs_reconstruct (real) (R3, maxEdges, nCells)

Units	unitless
Description	Coefficients to reconstruct velocity vectors at cell centers
Accessed in code	as coeffs_reconstruct from the mesh

cofrz (real) (nVertLevels, Time)

Units	s m ⁻¹
Description	coefficient for implicit contribution of Omega to density update
Accessed in code	as cofrz from the diag

coftz (real) (nVertLevelsP1, nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	s K
Description	coefficient for implicit contribution of omega vertical derivative to the theta_m update
Accessed in code	as coftz from the diag

cofwr (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	coefficient for implicit contribution of density to the vertical velocity update
Accessed in code	as cofwr from the diag

cofwt (real) (nVertLevels, nCells, Time)

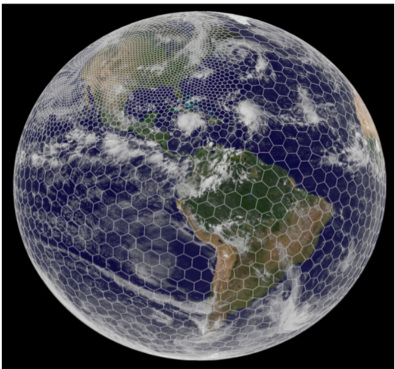
Units	m s ⁻¹ K ⁻¹
Description	coefficient for implicit contribution of density to the vertical velocity update
Accessed in code	as cofwt from the diag

cofwz (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹ K ⁻¹
Description	coefficient for implicit contribution of density to the vertical velocity update
Accessed in code	as cofwz from the diag

con (real) (nCells)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	orographic convexity
Accessed in code	as con from the sfc_input

coszr (real) (nCells, Time)

Units	unitless
Description	cosine of zenith solar angle
Accessed in code	as coszr from the diag_physics

cov (real) (nVertLevels, nCells, Time)

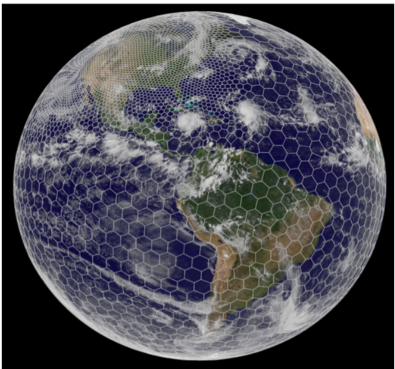
Units	K kg kg ⁻¹
Description	liquid water - liquid water potential temperature covariance
Allocated by	bl_mynn_in
Accessed in code	as cov from the diag_physics

cpm (real) (nCells, Time)

Units	J K ⁻¹ kg ⁻¹
Description	specific heat of dry air at constant pressure at lowest model level
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as cpm from the diag_physics

cqs2 (real) (nCells, Time)

Units	m s ⁻¹
Description	surface exchange coefficient for moisture at 2-meter
Allocated by	bl_mynn_in, bl_yсу_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as cqs2 from the diag_physics
---------------------	-------------------------------------

cqu (real) (nVertLevels, nEdges, Time)

Units	unitless
Description	rho_d/rho_m at cell edge (u points)
Accessed in code	as cqu from the diag

cqw (real) (nVertLevels, nCells, Time)

Units	unitless
Description	rho_d/rho_m at w points
Accessed in code	as cqw from the diag

cubot (real) (nCells, Time)

Units	unitless
Description	index of highest level of convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in
Accessed in code	as cubot from the diag_physics

cuprec (real) (nCells, Time)

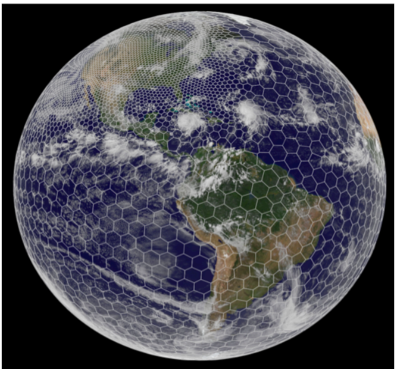
Units	mm s ⁻¹
Description	convective precipitation rate
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as cuprec from the diag_physics

cutop (real) (nCells, Time)

Units	unitless
Description	index of lowest level of convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in
Accessed in code	as cutop from the diag_physics

dcEdge (real) (nEdges)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Spherical distance between cells separated by an edge
Accessed in code	as dcEdge from the mesh

defc_a (real) (maxEdges, nCells)

Units	unitless
Description	Coefficients for computing the off-diagonal components of the horizontal deformation
Accessed in code	as defc_a from the mesh

defc_b (real) (maxEdges, nCells)

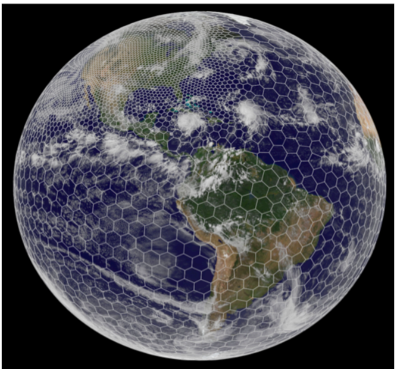
Units	unitless
Description	Coefficients for computing the diagonal components of the horizontal deformation
Accessed in code	as defc_b from the mesh

delta (real) (nCells, Time)

Units	m
Description	entrainment layer depth from PBL scheme
Allocated by	bl_yсу_in
Accessed in code	as delta from the diag_physics

depv_dt_bl (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
-------	---------------------



[MPAS Atmosphere](#)

Menu

Description	Diabatic EPV tendency from PBL
Accessed in code	as depv_dt_bl from the diag

depv_dt_cu (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency from convection
Accessed in code	as depv_dt_cu from the diag

depv_dt_diab (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Sum of calculated EPV tendencies from diabatic processes
Accessed in code	as depv_dt_diab from the diag

depv_dt_diab_pv (real) (nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency on dynamic tropopause
Accessed in code	as depv_dt_diab_pv from the diag

depv_dt_fric (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Sum of calculated EPV tendencies from frictional processes



[MPAS Atmosphere](#)

Menu

Accessed in code	as depv_dt_fric from the diag
---------------------	--

depv_dt_fric_pv (real) (nCells, Time)

Units	PVU s ⁻¹
Description	Frictional EPV tendency on dynamic tropopause
Accessed in code	as depv_dt_fric_pv from the diag

depv_dt_lw (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency from longwave radiation
Accessed in code	as depv_dt_lw from the diag

depv_dt_mix (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency from explicit numerical mixing
Accessed in code	as depv_dt_mix from the diag

depv_dt_mp (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency from microphysics
Accessed in code	as depv_dt_mp from the diag



[MPAS Atmosphere](#)

Menu

depv_dt_sw (real) (nVertLevels, nCells, Time)

Units	PVU s ⁻¹
Description	Diabatic EPV tendency from shortwave radiation
Accessed in code	as depv_dt_sw from the diag

deriv_two (real) (FIFTEEN, TWO, nEdges)

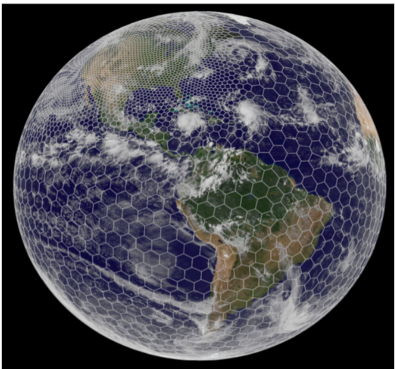
Units	unitless
Description	weights for cell-centered second derivative, normal to edge, for transport scheme
Accessed in code	as deriv_two from the mesh

det_qv (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	EDMF detrainment of water vapor mixing ratio in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as det_qv from the diag_physics

det_thl (real) (nVertLevels, nCells, Time)

Units	K s ⁻¹
Description	EDMF detrainment of liquid-ice potential temperature in the MYNN PBL scheme



[MPAS Atmosphere](#)

Menu

Allocated by	bl_mynn_in
Accessed in code	as det_thl from the diag_physics

dewpoint_100hPa (real) (nCells, Time)

Units	K
Description	Dewpoint temperature vertically interpolated to 100 hPa
Accessed in code	as dewpoint_100hPa from the diag

dewpoint_200hPa (real) (nCells, Time)

Units	K
Description	Dewpoint temperature vertically interpolated to 200 hPa
Accessed in code	as dewpoint_200hPa from the diag

dewpoint_250hPa (real) (nCells, Time)

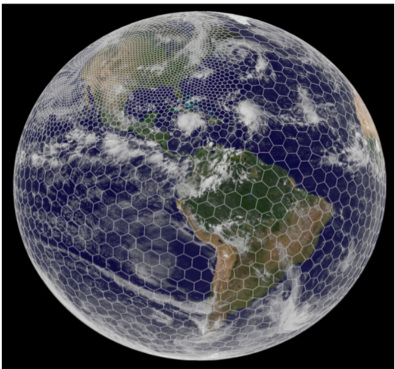
Units	K
Description	Dewpoint temperature vertically interpolated to 250 hPa
Accessed in code	as dewpoint_250hPa from the diag

dewpoint_500hPa (real) (nCells, Time)

Units	K
Description	Dewpoint temperature vertically interpolated to 500 hPa
Accessed in code	as dewpoint_500hPa from the diag

dewpoint_50hPa (real) (nCells, Time)

Units	K
-------	---



[MPAS Atmosphere](#)

Menu

Description	Dewpoint temperature vertically interpolated to 50 hPa
Accessed in code	as dewpoint_50hPa from the diag

dewpoint_700hPa (real) (nCells, Time)

Units	K
Description	Dewpoint temperature vertically interpolated to 700 hPa
Accessed in code	as dewpoint_700hPa from the diag

dewpoint_850hPa (real) (nCells, Time)

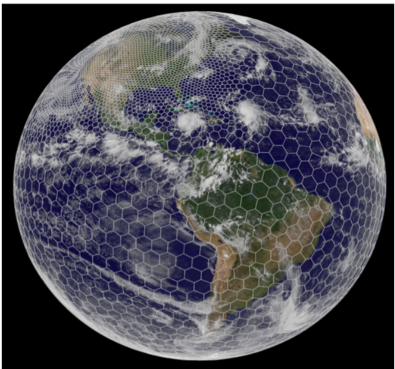
Units	K
Description	Dewpoint temperature vertically interpolated to 850 hPa
Accessed in code	as dewpoint_850hPa from the diag

dewpoint_925hPa (real) (nCells, Time)

Units	K
Description	Dewpoint temperature vertically interpolated to 925 hPa
Accessed in code	as dewpoint_925hPa from the diag

dewpoint_surface (real) (nCells, Time)

Units	K
Description	Dewpoint temperature at midpoint of lowest model layer
Accessed in code	as dewpoint_surface from the diag



[MPAS Atmosphere](#)

Menu

divergence (real) (nVertLevels, nCells, Time)

Units	s ⁻¹
Description	Horizontal velocity divergence at cell center
Accessed in code	as divergence from the diag

dqke (real) (nVertLevels, nCells, Time)

Units	m ² s ⁻²
Description	TKE change
Allocated by	bl_mynn_in
Accessed in code	as dqke from the diag_physics

dss (real) (nVertLevels, nCells)

Units	unitless
Description	w-damping coefficient
Accessed in code	as dss from the mesh

dtaux3d (real) (nVertLevels, nCells, Time)

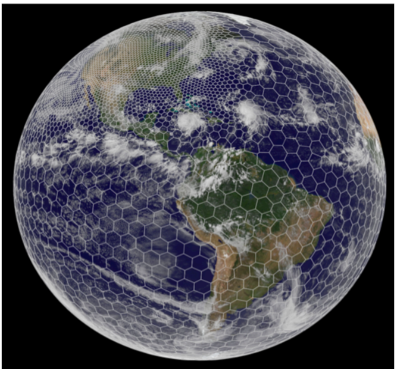
Units	m s ⁻¹
Description	gravity wave drag over orography u-stress
Accessed in code	as dtaux3d from the diag_physics

dtauy3d (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	gravity wave drag over orography v-stress
Accessed in code	as dtauy3d from the diag_physics

dtheta_dt_mix (real) (nVertLevels, nCells, Time)

Units	K s ⁻¹
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	Potential temperature heating rate from explicit numerical mixing
Accessed in code	as dtheta_dt_mix from the diag

dtheta_dt_mp (real) (nVertLevels, nCells, Time)

Units	K s^{-1}
Description	Potential temperature heating rate from microphysics
Accessed in code	as dtheta_dt_mp from the diag

dusfcg (real) (nCells, Time)

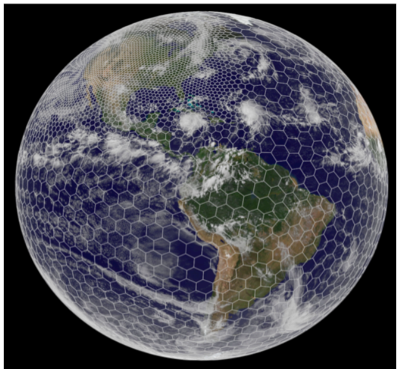
Units	Pa m s^{-1}
Description	vertically-integrated gravity wave drag over orography u-stress
Accessed in code	as dusfcg from the diag_physics

dvEdge (real) (nEdges)

Units	m
Description	Spherical distance between vertex endpoints of an edge
Accessed in code	as dvEdge from the mesh

dvsfcg (real) (nCells, Time)

Units	Pa m s^{-1}
Description	vertically-integrated gravity wave drag over orography v-stress



[MPAS Atmosphere](#)

Menu

Accessed in code	as dvsfcg from the diag_physics
---------------------	---------------------------------------

dzs (real) (nSoilLevels, nCells, Time)

Units	m
Description	soil layer thickness
Accessed in code	as dzs from the sfc_input

dzu (real) (nVertLevels)

Units	unitless
Description	d(zeta) at w levels
Accessed in code	as dzu from the mesh

east (real) (R3, nCells)

Units	unitless
Description	Cartesian components of the local unit vector pointing east
Accessed in code	as east from the mesh

edgeNormalVectors (real) (R3, nEdges)

Units	unitless
Description	Cartesian components of the vector normal to an edge and tangential to the surface of the sphere
Accessed in code	as edgeNormalVectors from the mesh

edgesOnCell (integer) (maxEdges, nCells)

Units	•
Description	IDs of edges forming the boundary of a cell



[MPAS Atmosphere](#)

Menu

Accessed in code	as edgesOnCell from the mesh
---------------------	---------------------------------------

edgesOnCell_sign (real) (maxEdges, nCells)

Units	•
Description	Sign for edges surrounding a cell: positive for positive outward normal velocity
Accessed in code	as edgesOnCell_sign from the mesh

edgesOnEdge (integer) (maxEdges2, nEdges)

Units	•
Description	IDs of edges involved in reconstruction of tangential velocity for an edge
Accessed in code	as edgesOnEdge from the mesh

edgesOnVertex (integer) (vertexDegree, nVertices)

Units	•
Description	IDs of the edges that meet at a vertex
Accessed in code	as edgesOnVertex from the mesh

edgesOnVertex_sign (real) (vertexDegree, nVertices)

Units	•
Description	Sign for edges incident with a vertex: positive for positive inward tengential velocity
Accessed in code	as edgesOnVertex_sign from the mesh

edmf_a (real) (nVertLevels, nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	EDMF relative updraft area of moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as edmf_a from the diag_physics

edmf_ent (real) (nVertLevels, nCells, Time)

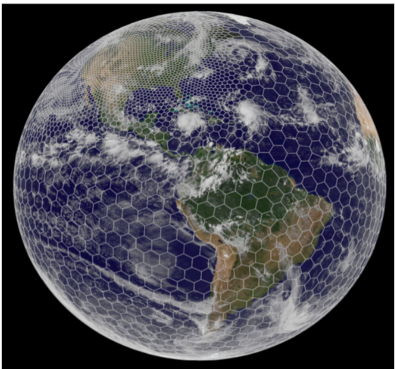
Units	m ⁻¹
Description	EDMF entrainment rate in moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as edmf_ent from the diag_physics

edmf_qc (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹
Description	EDMF liquid water mixing ratio in moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as edmf_qc from the diag_physics

edmf_qt (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹
Description	EDMF total water mixing ratio in moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as edmf_qt from the diag_physics
------------------	----------------------------------

edmf_thl (real) (nVertLevels, nCells, Time)

Units	K
Description	EDMF liquid-ice water potential temperature in moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as edmf_thl from the diag_physics

edmf_w (real) (nVertLevels, nCells, Time)

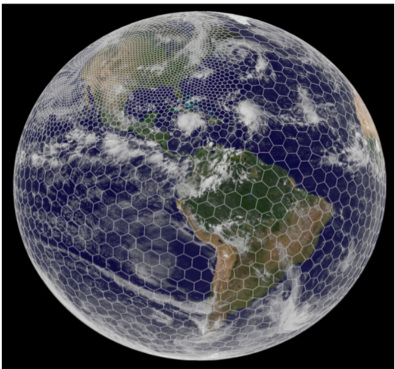
Units	m s ⁻¹
Description	EDMF vertical velocity of moist updrafts in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as edmf_w from the diag_physics

el_pbl (real) (nVertLevels, nCells, Time)

Units	m
Description	mixing length from PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as el_pbl from the diag_physics

emstot (real) (nVertLevelsP1, nCells, Time)

Units	•
Description	Total emissivity
Accessed in code	as emstot from the diag_physics



ertel_pv (real) (nVertLevels, nCells, Time)

Units	PVU
Description	Ertel’s potential vorticity
Accessed in code	as ertel_pv from the diag

euler_tend_theta (real) (nVertLevels, nCells, Time)

Units	$\text{kg K m}^{-3} \text{s}^{-1}$
Description	tendency of coupled potential temperature $\rho \cdot \theta_m / \text{zz}$ from dynamics and physics that does not change over a Runge-Kutta timestep (it excludes the flux divergence)
Accessed in code	as theta_euler from the tend

euler_tend_u (real) (nVertLevels, nEdges, Time)

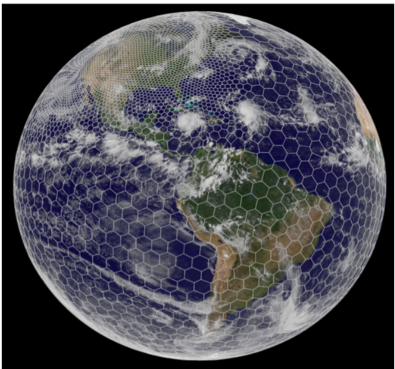
Units	m s^{-2}
Description	Tendency of u from dynamics
Accessed in code	as u_euler from the tend

euler_tend_w (real) (nVertLevelsP1, nCells, Time)

Units	m s^{-2}
Description	Tendency of w from dynamics
Accessed in code	as w_euler from the tend

evapprod (real) (nVertLevels, nCells, Time)

Units	s^{-1}
-------	-----------------



[MPAS Atmosphere](#)

Menu

Description	rain evaporation rate
Allocated by	mp_thompson_in
Accessed in code	as evapprod from the diag_physics

exch_h (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-1}$
Description	exchange coefficient
Allocated by	bl_yсу_in
Accessed in code	as exch_h from the diag_physics

exner (real) (nVertLevels, nCells, Time)

Units	unitless
Description	Exner function
Accessed in code	as exner from the diag

exner_base (real) (nVertLevels, nCells, Time)

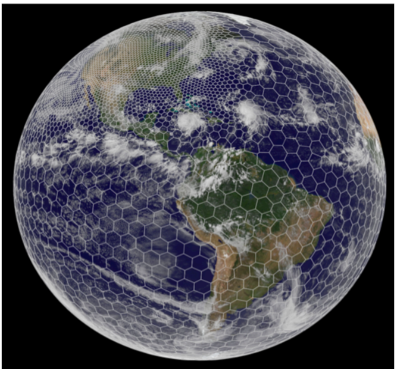
Units	unitless
Description	Base-state Exner function
Accessed in code	as exner_base from the diag

fEdge (real) (nEdges)

Units	unitless
Description	Coriolis parameter at an edge
Accessed in code	as fEdge from the mesh

fh (real) (nCells, Time)

Units	unitless
Description	integrated stability function for heat
Allocated by	bl_yсу_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as fh from the diag_physics
------------------	-----------------------------

flhc (real) (nCells, Time)

Units	W m ⁻² K ⁻¹
Description	exchange coefficient for heat
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as flhc from the diag_physics

flqc (real) (nCells, Time)

Units	kg m ⁻² s ⁻¹
Description	exchange coefficient for moisture
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as flqc from the diag_physics

fm (real) (nCells, Time)

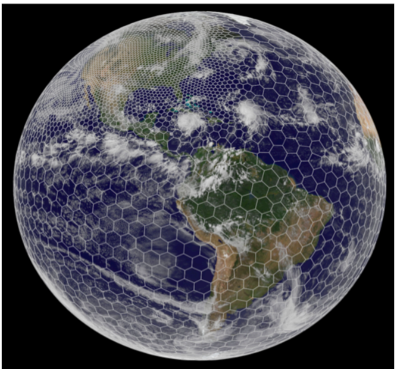
Units	unitless
Description	integrated stability function for moisture
Allocated by	bl_yсу_in
Accessed in code	as fm from the diag_physics

fVertex (real) (nVertices)

Units	unitless
Description	Coriolis parameter at a vertex
Accessed in code	as fVertex from the mesh

fzm (real) (nVertLevels)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	Weight for linear interpolation to w(k) point for u(k) level variable
Accessed in code	as fzm from the mesh

fzp (real) (nVertLevels)

Units	unitless
Description	Weight for linear interpolation to w(k) point for u(k-1`) level variable
Accessed in code	as fzp from the mesh

gamma_tri (real) (nVertLevels, nCells, Time)

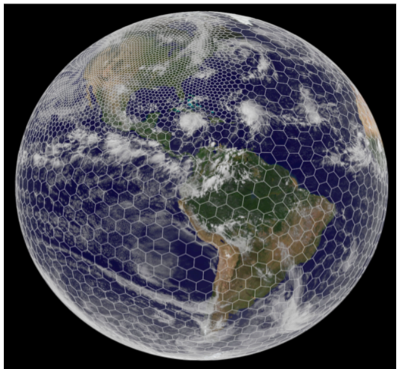
Units	unitless
Description	implicit tridiagonal solve coefficients
Accessed in code	as gamma_tri from the diag

glw (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky downward surface longwave radiation
Accessed in code	as glw from the diag_physics

gradPVn (real) (nVertLevels, nEdges, Time)

Units	???
Description	Gradient in the normal direction of PV at edge locations
Accessed in code	as gradPVn from the diag



[MPAS Atmosphere](#)

Menu

gradPVt (real) (nVertLevels, nEdges, Time)

Units	???
Description	Gradient in the tangential direction of PV at edge locations
Accessed in code	as gradPVt from the diag

graupelnc (real) (nCells, Time)

Units	mm
Description	accumulated grid-scale precipitation of graupel
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as graupelnc from the diag_physics

graupelncv (real) (nCells, Time)

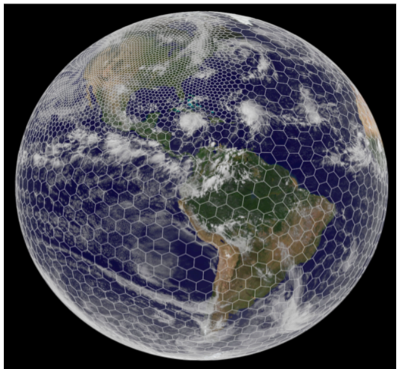
Units	mm
Description	time-step grid-scale precipitation of graupel
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as graupelncv from the diag_physics

grdflx (real) (nCells, Time)

Units	W m ⁻²
Description	ground heat flux
Accessed in code	as grdflx from the diag_physics

greenfrac (real) (nMonths, nCells)

Units	unitless
Description	monthly-mean climatological greeness fraction
Accessed in code	as greenfrac from the sfc_input



[MPAS Atmosphere](#)

Menu

gsw (real) (nCells, Time)

Units	W m ⁻²
Description	net surface shortwave radiation flux
Accessed in code	as gsw from the diag_physics

gz1oz0 (real) (nCells, Time)

Units	unitless
Description	log(z/z0) where z0 is roughness length
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as gz1oz0 from the diag_physics

h_divergence (real) (nVertLevels, nCells, Time)

Units	???
Description	???
Accessed in code	as h_divergence from the diag

h_oml (real) (nCells, Time)

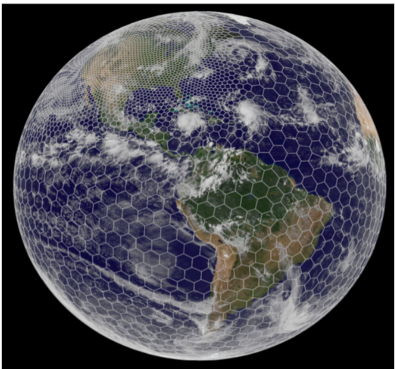
Units	m
Description	ocean mixed layer depth
Accessed in code	as h_oml from the diag_physics

h_oml_initial (real) (nCells, Time)

Units	m
Description	ocean mixed layer depth at initial time
Accessed in code	as h_oml_initial from the diag_physics

height_100hPa (real) (nCells, Time)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Geometric height interpolated to 100 hPa
Accessed in code	as height_100hPa from the diag

height_200hPa (real) (nCells, Time)

Units	m
Description	Geometric height interpolated to 200 hPa
Accessed in code	as height_200hPa from the diag

height_250hPa (real) (nCells, Time)

Units	m
Description	Geometric height interpolated to 250 hPa
Accessed in code	as height_250hPa from the diag

height_500hPa (real) (nCells, Time)

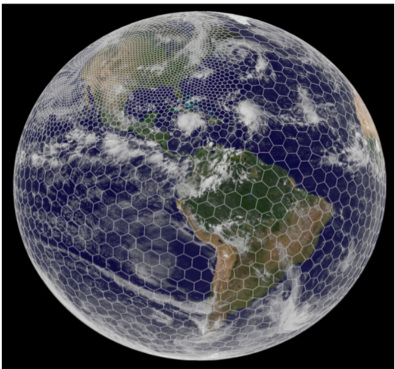
Units	m
Description	Geometric height interpolated to 500 hPa
Accessed in code	as height_500hPa from the diag

height_50hPa (real) (nCells, Time)

Units	m
Description	Geometric height interpolated to 50 hPa
Accessed in code	as height_50hPa from the diag

height_700hPa (real) (nCells, Time)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Geometric height interpolated to 700 hPa
Accessed in code	as height_700hPa from the diag

height_850hPa (real) (nCells, Time)

Units	m
Description	Geometric height interpolated to 850 hPa
Accessed in code	as height_850hPa from the diag

height_925hPa (real) (nCells, Time)

Units	m
Description	Geometric height interpolated to 925 hPa
Accessed in code	as height_925hPa from the diag

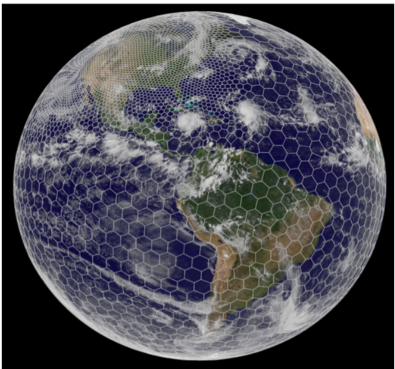
hfx (real) (nCells, Time)

Units	W m ⁻²
Description	upward heat flux at the surface
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as hfx from the diag_physics

hpbl (real) (nCells, Time)

Units	m
Description	Planetary Boundary Layer (PBL) height
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as hpbl from the diag_physics

hu_oml (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	$\text{m}^2 \text{s}^{-1}$
Description	ocean mixed layer integrated u (zonal velocity)
Accessed in code	as hu_oml from the diag_physics

hv_oml (real) (nCells, Time)

Units	$\text{m}^2 \text{s}^{-1}$
Description	ocean mixed layer integrated v (meridional velocity)
Accessed in code	as hv_oml from the diag_physics

i_aclwdnb (integer) (nCells, Time)

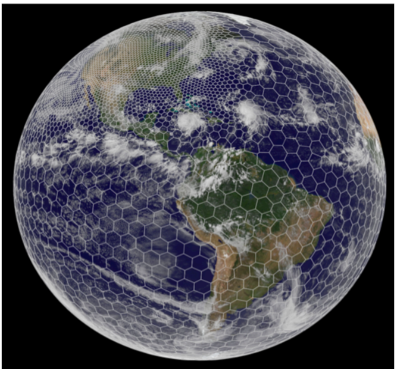
Units	unitless
Description	incidence of accumulated all-sky downward surface longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwdnb from the diag_physics

i_aclwdnbc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated clear-sky downward surface longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwdnbc from the diag_physics

i_aclwdnt (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky downward top-of-atmosphere longwave radiation greater than config_bucket_radt



[MPAS Atmosphere](#)

Menu

Accessed in code	as i_aclwdnt from the diag_physics
------------------	------------------------------------

i_aclwdntc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated clear-sky downward top-of-atmosphere longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwdntc from the diag_physics

i_aclwupb (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky upward surface longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwupb from the diag_physics

i_aclwupbc (integer) (nCells, Time)

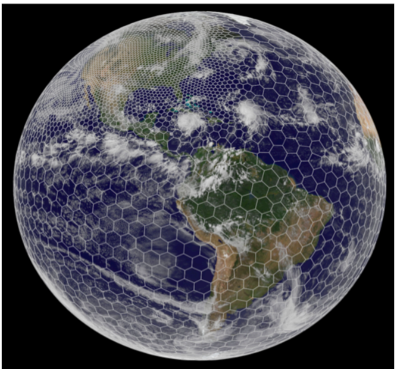
Units	unitless
Description	incidence of accumulated clear-sky upward surface longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwupbc from the diag_physics

i_aclwupt (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky upward top-of-atmosphere longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwupt from the diag_physics

i_aclwuptc (integer) (nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	incidence of accumulated clear-sky upward top-of-atmosphere longwave radiation greater than config_bucket_radt
Accessed in code	as i_aclwuptc from the diag_physics

i_acswdnb (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky downward surface shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswdnb from the diag_physics

i_acswdnbc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated clear-sky downward surface shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswdnbc from the diag_physics

i_acswdnt (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky downward top-of-atmosphere shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswdnt from the diag_physics

i_acswdnrtc (integer) (nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	incidence of accumulated clear-sky downward top-of-atmosphere shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswdntc from the diag_physics

i_acswupb (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky upward surface shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswupb from the diag_physics

i_acswupbc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated clear-sky upward surface shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswupbc from the diag_physics

i_acswupt (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated all-sky upward top-of-atmosphere shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswupt from the diag_physics

i_acswuptc (integer) (nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	incidence of accumulated clear-sky upward top-of-atmosphere shortwave radiation greater than config_bucket_radt
Accessed in code	as i_acswuptc from the diag_physics

i_rainc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated convective precipitation greater than config_bucket_rainc
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as i_rainc from the diag_physics

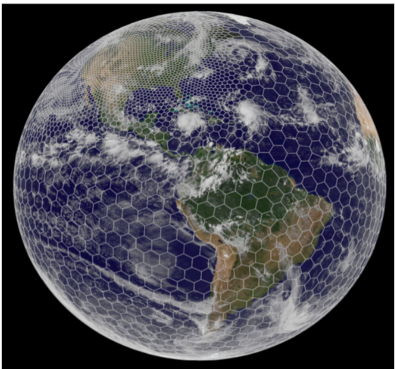
i_rainnc (integer) (nCells, Time)

Units	unitless
Description	incidence of accumulated grid-scale precipitation greater than config_bucket_rainnc
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in
Accessed in code	as i_rainnc from the diag_physics



iLev_DT (integer) (nCells, Time)

Units	•
Description	Lowest vertical level at or above dynamic tropopause (.lt.1 if 2 PVU below column; .gt.nLevels if 2PVU above column)



[MPAS Atmosphere](#)

Menu

Accessed in code	as iLev_DT from the diag
------------------	--------------------------

indexToCellID (integer) (nCells)

Units	•
Description	Mapping from local array index to global cell ID
Accessed in code	as indexToCellID from the mesh

indexToEdgeID (integer) (nEdges)

Units	•
Description	Mapping from local array index to global edge ID
Accessed in code	as indexToEdgeID from the mesh

indexToVertexID (integer) (nVertices)

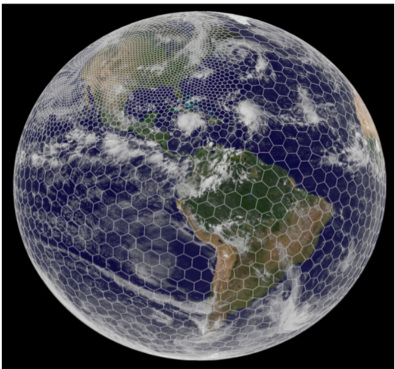
Units	•
Description	Mapping from local array index to global vertex ID
Accessed in code	as indexToVertexID from the mesh

initial_time (text) ()

Units	YYYY-MM-DD_hh:mm:ss
Description	Model initialization time
Accessed in code	as initial_time from the state

inTropo (integer) (nVertLevels, nCells, Time)

Units	0/1
Description	1 if within troposphere based on EPV flood fill



[MPAS Atmosphere](#)

Menu

Accessed in code	as inTropo from the diag
---------------------	--------------------------------

invAreaCell (real) (nCells)

Units	m ⁻²
Description	Inverse of Voronoi cell area
Accessed in code	as invAreaCell from the mesh

invAreaTriangle (real) (nVertices)

Units	m ⁻²
Description	Inverse area of a Delaunay triangle
Accessed in code	as invAreaTriangle from the mesh

invDcEdge (real) (nEdges)

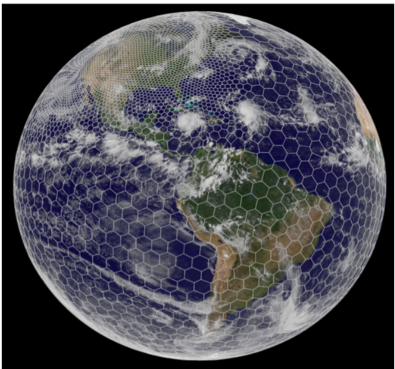
Units	m ⁻¹
Description	Inverse distance between cells separated by an edge
Accessed in code	as invDcEdge from the mesh

invDvEdge (real) (nEdges)

Units	m ⁻¹
Description	Inverse distance between vertex endpoints of an edge
Accessed in code	as invDvEdge from the mesh

isice_lu (integer) ()

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	Index category for snow/ice
Accessed in code	as isice from the sfc_input

isltyp (integer) (nCells)

Units	unitless
Description	dominant soil category
Accessed in code	as isltyp from the sfc_input

iswater_lu (integer) ()

Units	unitless
Description	Index category for water
Accessed in code	as iswater from the sfc_input

ivgtyp (integer) (nCells)

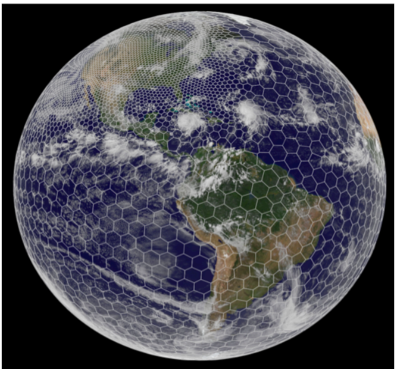
Units	unitless
Description	dominant vegetation category
Accessed in code	as ivgtyp from the sfc_input

k22_shallow (integer) (nCells, Time)

Units	unitless
Description	index of convection at k22 level in Grell-Freitas convection scheme
Allocated by	cu_grell_freitas_in
Accessed in code	as k22_shallow from the diag_physics

kbcon_shallow (integer) (nCells, Time)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	index of lowest level of shallow convection in Grell-Freitas convection scheme
Allocated by	cu_grell_freitas_in
Accessed in code	as kbcon_shallow from the diag_physics

kdif (real) (nVertLevels, nCells, Time)

Units	m ² s ⁻¹
Description	Smagorinsky horizontal eddy viscosity
Accessed in code	as kdif from the diag

ke (real) (nVertLevels, nCells, Time)

Units	m ² s ⁻²
Description	Kinetic energy at a cell center
Accessed in code	as ke from the diag

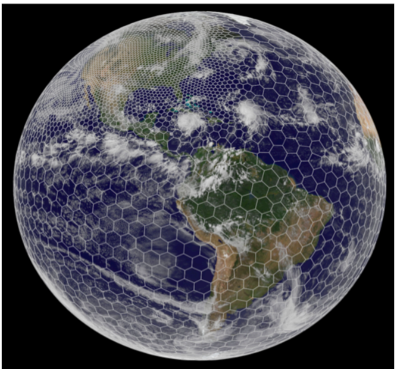
kiteAreasOnVertex (real) (vertexDegree, nVertices)

Units	m ²
Description	Intersection areas between primal (Voronoi) and dual (triangular) mesh cells
Accessed in code	as kiteAreasOnVertex from the mesh

kiteForCell (integer) (maxEdges, nCells)

Units	•
Description	Index of kite in kiteAreasOnVertex that lies within a cell for each of verticesOnCell
Accessed in code	as kiteForCell from the mesh

kpbl (integer) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	unitless
Description	index level of PBL top
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as kpbl from the diag_physics

ktop_deep (integer) (nCells, Time)

Units	unitless
Description	index of highest level of deep convection in Grell-Freitas convection scheme
Allocated by	cu_grell_freitas_in
Accessed in code	as ktop_deep from the diag_physics

ktop_shallow (integer) (nCells, Time)

Units	unitless
Description	index of highest level of shallow convection in Grell-Freitas convection scheme
Allocated by	cu_grell_freitas_in
Accessed in code	as ktop_shallow from the diag_physics

kzh (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-1}$
Description	vertical diffusion coefficient of potential temperature
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as kzh from the diag_physics

kzm (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-1}$
-------	----------------------------



[MPAS Atmosphere](#)

Menu

Description	vertical diffusion coefficient of mommentum
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as kzm from the diag_physics

kzq (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-1}$
Description	vertical diffusion coefficient of water vapor and cloud condensates
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as kzq from the diag_physics

lai (real) (nCells, Time)

Units	$\text{m}^{-2} \text{m}^{-2}$
Description	leaf area index
Accessed in code	as lai from the diag_physics

landmask (integer) (nCells)

Units	unitless
Description	land-ocean mask (1=land ; 0=ocean)
Accessed in code	as landmask from the sfc_input

latCell (real) (nCells)

Units	rad
Description	Latitude of cells
Accessed in code	as latCell from the mesh

latEdge (real) (nEdges)



[MPAS Atmosphere](#)

Menu

Units	rad
Description	Latitude of edges
Accessed in code	as latEdge from the mesh

latVertex (real) (nVertices)

Units	rad
Description	Latitude of vertices
Accessed in code	as latVertex from the mesh

lbc_rho (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³ s ⁻¹
Description	Lateral boundary tendency of rho
Accessed in code	as lbc_rho from the lbc

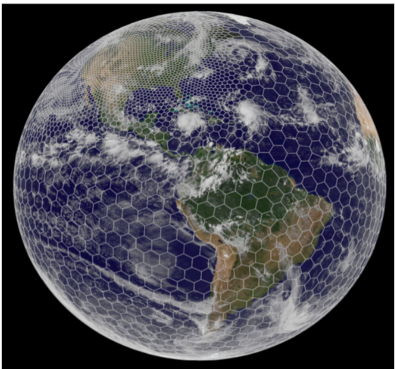
lbc_rho_edge (real) (nVertLevels, nEdges, Time)

Units	kg m ⁻³ s ⁻¹
Description	Lateral boundary tendency of rho_zz averaged from cell centers to cell faces (edges)
Accessed in code	as lbc_rho_edge from the lbc

lbc_rho_zz (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³ s ⁻¹
Description	Lateral boundary tendency of rho_zz
Accessed in code	as lbc_rho_zz from the lbc

lbc_rtheta_m (real) (nVertLevels, nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	$\text{kg K m}^{-3} \text{s}^{-2}$
Description	Lateral boundary tendency of rho_zz-coupled theta_m
Accessed in code	as lbc_rtheta_m from the lbc

lbc_ru (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{s}^{-2}$
Description	Lateral boundary tendency of rho_zz-coupled u
Accessed in code	as lbc_ru from the lbc

lbc_scalars (real) (nVertLevels, nCells, Time)

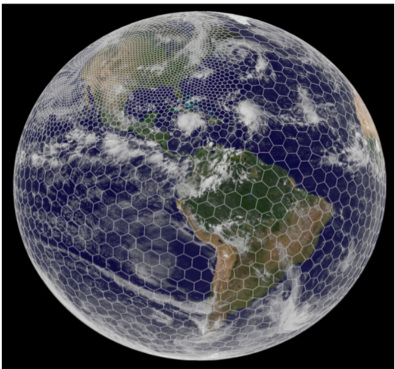
Description	lbc_scalars is a variable array used in code which is made up of the constituent fields listed below.
Accessed in code	lbc_scalars from the lbc pool

lbc_qv – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of water vapor mixing ratio
Units	$\text{kg kg}^{-1} \text{s}^{-1}$
Array Group	moist

lbc_qc – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of cloud water mixing ratio
Units	$\text{kg kg}^{-1} \text{s}^{-1}$



[MPAS Atmosphere](#)

Menu

Array Group	moist
Allocated by	bl_mynn_in, cu_ntiedtke_in, mp_kessler_in, mp_thompson_in, mp_wsm6_in

lbc_qr – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of rain water mixing ratio
Units	kg kg ⁻¹ s ⁻¹
Array Group	moist
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in

lbc_qi – constituent field of var_array
lbc_scalars

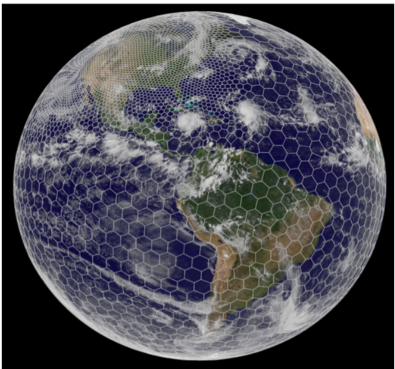
Description	Lateral boundary tendency of ice mixing ratio
Units	kg kg ⁻¹ s ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, cu_ntiedtke_in, mp_thompson_in, mp_wsm6_in

lbc_qs – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of snow mixing ratio
Units	kg kg ⁻¹ s ⁻¹
Array Group	moist
Allocated by	mp_thompson_in, mp_wsm6_in

lbc_qg – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of graupel mixing ratio
Units	kg kg ⁻¹ s ⁻¹
Array Group	moist



[MPAS Atmosphere](#)

Menu

Allocated by	mp_thompson_in, mp_wsm6_in
--------------	----------------------------

lbc_nr – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of rain number concentration
Units	m ⁻³ s ⁻¹
Array Group	number
Allocated by	mp_thompson_in

lbc_ni – constituent field of var_array
lbc_scalars

Description	Lateral boundary tendency of ice number concentration
Units	m ⁻³ s ⁻¹
Array Group	number
Allocated by	bl_mynn_in, mp_thompson_in

lbc_theta (real) (nVertLevels, nCells, Time)

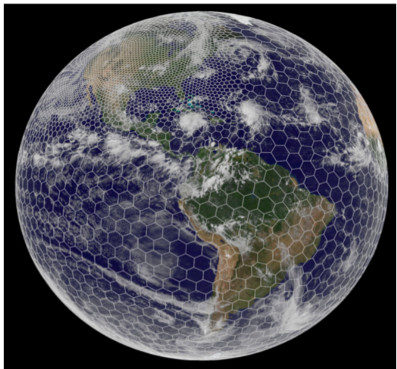
Units	K s ⁻²
Description	Lateral boundary tendency of theta
Accessed in code	as lbc_theta from the lbc

lbc_u (real) (nVertLevels, nEdges, Time)

Units	m s ⁻²
Description	Lateral boundary tendency of u
Accessed in code	as lbc_u from the lbc

lbc_w (real) (nVertLevelsP1, nCells, Time)

Units	m s ⁻²
Description	Lateral boundary tendency of w



[MPAS Atmosphere](#)

Menu

Accessed in code	as lbc_w from the lbc
---------------------	--------------------------

lcl (real) (nCells, Time)

Units	m
Description	Lifted condensation level
Accessed in code	as lcl from the diag

lfc (real) (nCells, Time)

Units	m
Description	Level of free convection
Accessed in code	as lfc from the diag

lh (real) (nCells, Time)

Units	W m ⁻²
Description	latent heat flux at the surface
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as lh from the diag_physics

localVerticalUnitVectors (real) (R3, nCells)

Units	unitless
Description	Cartesian components of the vector pointing in the local vertical direction for a cell
Accessed in code	as localVerticalUnitVectors from the mesh

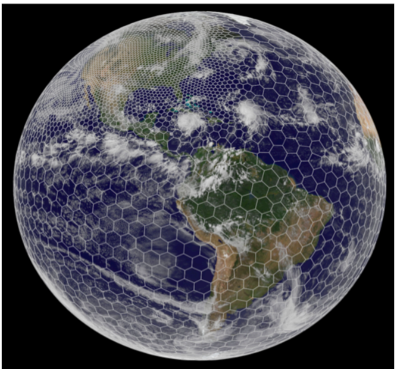


lonCell (real) (nCells)

Units	rad
Description	Longitude of cells
Accessed in code	as lonCell from the mesh

lonEdge (real) (nEdges)

Units	rad
-------	-----



[MPAS Atmosphere](#)

Menu

Description	Longitude of edges
Accessed in code	as lonEdge from the mesh

lonVertex (real) (nVertices)

Units	rad
Description	Longitude of vertices
Accessed in code	as lonVertex from the mesh

lwcf (real) (nCells, Time)

Units	W m ⁻²
Description	top-of-atmosphere cloud longwave radiative forcing
Accessed in code	as lwcf from the diag_physics

lwdnb (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky downward surface longwave radiation flux
Accessed in code	as lwdnb from the diag_physics

lwdnbc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky downward surface longwave radiation flux
Accessed in code	as lwdnbc from the diag_physics

lwdnt (real) (nCells, Time)

Units	W m ⁻²
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	all-sky downward top-of-the-atmosphere longwave radiation flux
Accessed in code	as lwdnt from the diag_physics

lwdntc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky downward top-of-the-atmosphere longwave radiation flux
Accessed in code	as lwdntc from the diag_physics

lwupb (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky upward surface longwave radiation flux
Accessed in code	as lwupb from the diag_physics

lwupbc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky upward surface longwave radiation flux
Accessed in code	as lwupbc from the diag_physics

lwupt (real) (nCells, Time)

Units	W m ⁻²
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	all-sky upward top- of-the- atmosphere longwave radiation flux
Accessed in code	as lwupt from the diag_physics

lwuptc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky upward top- of-the- atmosphere longwave radiation flux
Accessed in code	as lwuptc from the diag_physics

m_hybi (real) (nAerLevels, nCells)

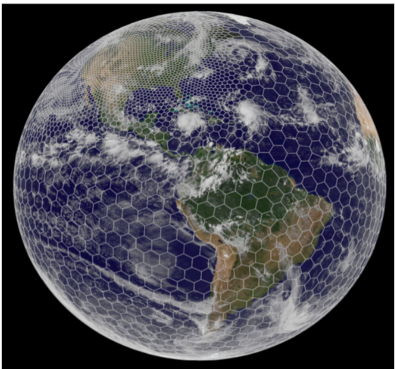
Units	unitless
Description	Matched hybi (needs to be re- checked)
Accessed in code	as m_hybi from the diag_physics

m_ps (real) (nCells, Time)

Units	Pa
Description	Surface pressure from match on MPAS grid
Accessed in code	as m_ps from the diag_physics

mavail (real) (nCells, Time)

Units	unitless
Description	surface moisture availability (between 0 and 1)
Allocated by	bl_mynn_in, bl_yсу_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as mavail from the diag_physics
------------------	---------------------------------

meanT_500_300 (real) (nCells, Time)

Units	K
Description	Mean temperature in the 300 hPa - 500 hPa layer
Accessed in code	as meanT_500_300 from the diag

meshDensity (real) (nCells)

Units	unitless
Description	Mesh density function (used when generating the mesh) evaluated at a cell
Accessed in code	as meshDensity from the mesh

meshScalingDel2 (real) (nEdges)

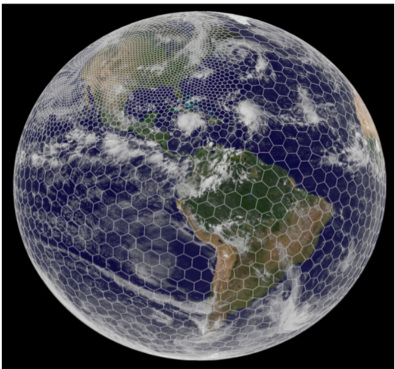
Units	unitless
Description	Scaling coefficient for ∇^2 eddy viscosity
Accessed in code	as meshScalingDel2 from the mesh

meshScalingDel4 (real) (nEdges)

Units	unitless
Description	Scaling coefficient for ∇^4 eddy hyper-viscosity
Accessed in code	as meshScalingDel4 from the mesh

meshScalingRegionalCell (real) (nCells)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	Cell-centered Scaling coefficient for relaxation zone
Accessed in code	as meshScalingRegionalCel from the mesh

meshScalingRegionalEdge (real) (nEdges)

Units	unitless
Description	Edge-centered Scaling coefficient for relaxation zone
Accessed in code	as meshScalingRegionalEd from the mesh

mminlu (text) ()

Units	unitless
Description	land use classification
Accessed in code	as mminlu from the sfc_input

mol (real) (nCells, Time)

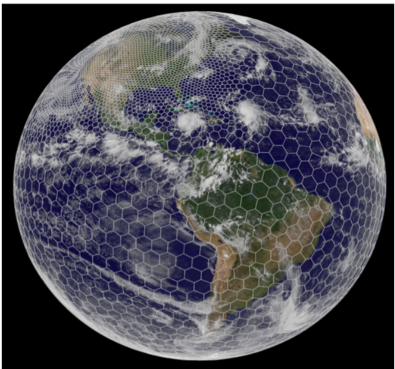
Units	K
Description	T* in similarity theory
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as mol from the diag_physics

mslp (real) (nCells, Time)

Units	Pa
Description	Mean sea-level pressure
Accessed in code	as mslp from the diag

mu_c (real) (nCells)

Units	unitless
Description	gamma shape parameter
Allocated by	mp_thompson_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as mu_c from the diag_physics
------------------	-------------------------------

nAdvCellsForEdge (integer) (nEdges)

Units	•
Description	Number of cells used to reconstruct a cell-based field at an edge
Accessed in code	as nAdvCellsForEdge from the mesh

nca (real) (nCells, Time)

Units	s
Description	relaxation time for KF parameterization of convection
Allocated by	cu_kain_fritsch_in
Accessed in code	as nca from the diag_physics

ndays_accum (real) (nCells, Time)

Units	unitless
Description	number of accumulated days in a year
Accessed in code	as ndays_accum from the diag_physics

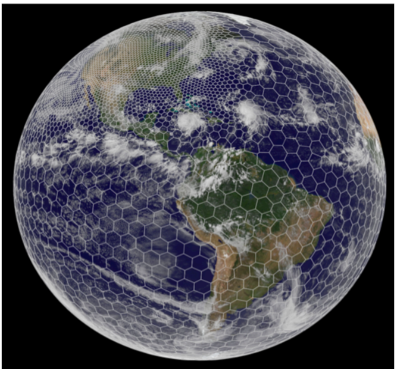
nearestRelaxationCell (integer) (nCells)

Units	
Description	For cells in the specified zone, gives the index of the nearest cell in the relaxation zone
Accessed in code	as nearestRelaxationCell from the mesh



nEdgesOnCell (integer) (nCells)

Units	•
-------	---



[MPAS Atmosphere](#)

Menu

Description	Number of edges forming the boundary of a cell
Accessed in code	as nEdgesOnCell from the mesh

nEdgesOnEdge (integer) (nEdges)

Units	•
Description	Number of edges involved in reconstruction of tangential velocity for an edge
Accessed in code	as nEdgesOnEdge from the mesh

nlrad (integer) (nCells, Time)

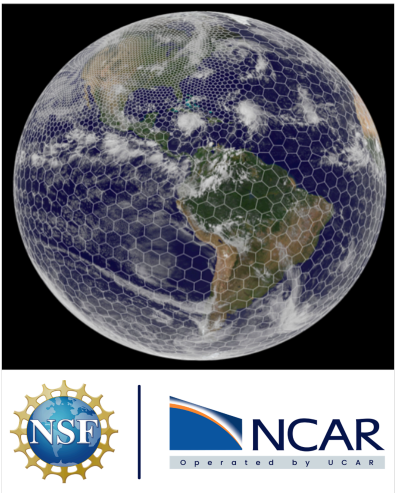
Units	•
Description	number of layers added above the model-top in the RRTMG lw radiation code
Accessed in code	as nlrad from the diag_physics

noahres (real) (nCells, Time)

Units	W m ⁻²
Description	residual of the Noah surface energy budget
Accessed in code	as noahres from the diag_physics

nominalMinDc (real) ()

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Nominal minimum dcEdge value where meshDensity == 1.0
Accessed in code	as nominalMinDc from the mesh

north (real) (R3, nCells)

Units	unitless
Description	Cartesian components of the unit vector pointing north
Accessed in code	as north from the mesh

nsteps_accum (real) (nCells, Time)

Units	unitless
Description	number of accumulated time-steps in a day
Accessed in code	as nsteps_accum from the diag_physics

nt_c (real) (nCells)

Units	nb kg ⁻¹
Description	one-moment constant cloud water droplet concentration
Allocated by	mp_thompson_in
Accessed in code	as nt_c from the diag_physics

o3clim (real) (nOznLevels, nCells, Time)

Units	mol mol ⁻¹
Description	climatological ozone on prescribed pressure levels at current time



[MPAS Atmosphere](#)

Menu

Accessed in code	as o3clim from the diag_physics
---------------------	---------------------------------------

o3vmr (real) (nVertLevels, nCells, Time)

Units	mol mol ⁻¹
Description	ozone volume mixing ratio
Accessed in code	as o3vmr from the diag_physics

oa1 (real) (nCells)

Units	unitless
Description	asymmetry of subgrid- scale orography for westerly flow
Accessed in code	as oa1 from the sfc_input

oa2 (real) (nCells)

Units	unitless
Description	asymmetry of subgrid- scale orography for southerly flow
Accessed in code	as oa2 from the sfc_input

oa3 (real) (nCells)

Units	unitless
Description	asymmetry of subgrid- scale orography for south- westerly flow
Accessed in code	as oa3 from the sfc_input

oa4 (real) (nCells)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	asymmetry of subgrid-scale orography for north-westerly flow
Accessed in code	as oa4 from the sfc_input

ol1 (real) (nCells)

Units	unitless
Description	effective orographic length for westerly flow
Accessed in code	as ol1 from the sfc_input

ol2 (real) (nCells)

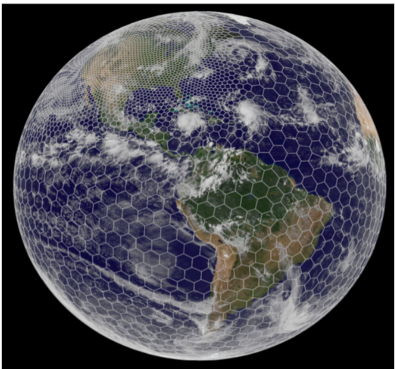
Units	unitless
Description	effective orographic length for southerly flow
Accessed in code	as ol2 from the sfc_input

ol3 (real) (nCells)

Units	unitles
Description	effective orographic length for south-westerly flow
Accessed in code	as ol3 from the sfc_input

ol4 (real) (nCells)

Units	unitless
Description	effective orographic length for north-westerly flow



[MPAS Atmosphere](#)

Menu

Accessed in code	as ol4 from the sfc_input
------------------	---------------------------

olrtoa (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky top-of-atmosphere outgoing longwave radiation flux
Accessed in code	as olrtoa from the diag_physics

ozmixm (real) (nMonths, nOznLevels, nCells)

Units	mol mol ⁻¹
Description	monthly-mean climatological ozone defined at fixed pressure levels
Accessed in code	as ozmixm from the atm_input

pin (real) (nOznLevels)

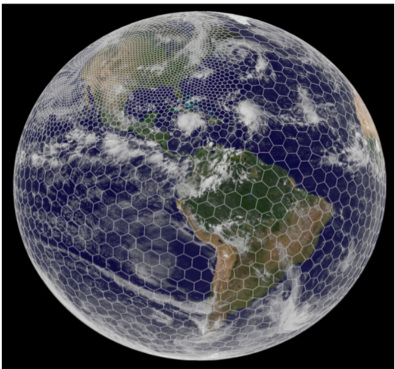
Units	Pa
Description	fixed pressure levels at which climatological ozone is defined
Accessed in code	as pin from the atm_input

plrad (real) (nCells, Time)

Units	Pa
Description	pressure at model-top
Accessed in code	as plrad from the diag_physics

potevp (real) (nCells, Time)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	accumulated potential evaporation
Accessed in code	as potevp from the diag_physics

precipw (real) (nCells, Time)

Units	kg m ⁻²
Description	precipitable water
Accessed in code	as precipw from the diag_physics

pressure (real) (nVertLevels, nCells, Time)

Units	Pa
Description	Pressure
Accessed in code	as pressure from the diag

pressure_base (real) (nVertLevels, nCells, Time)

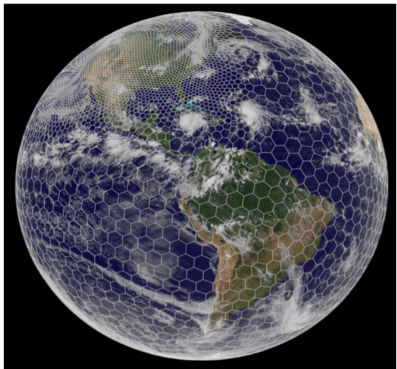
Units	Pa
Description	Base state pressure
Accessed in code	as pressure_base from the diag

pressure_p (real) (nVertLevels, nCells, Time)

Units	Pa
Description	Perturbation pressure
Accessed in code	as pressure_p from the diag

psih (real) (nCells, Time)

Units	unitless
Description	similarity stability function for heat
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as psih from the diag_physics



[MPAS Atmosphere](#)

Menu

psim (real) (nCells, Time)

Units	unitless
Description	similarity stability function for momentum
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as psim from the diag_physics

pv_cell (real) (nVertLevels, nCells, Time)

Units	s ⁻¹
Description	absolute vertical vorticity averaged to the cell center from the vertices
Accessed in code	as pv_cell from the diag

pv_edge (real) (nVertLevels, nEdges, Time)

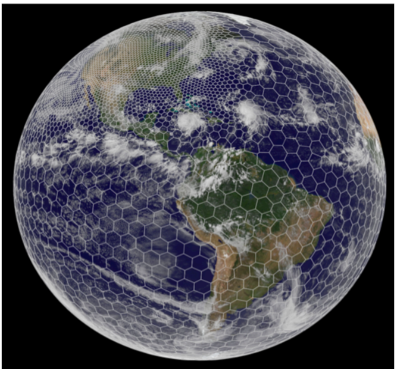
Units	s ⁻¹
Description	absolute vertical vorticity averaged to the cell edge from the vertices
Accessed in code	as pv_edge from the diag

pv_vertex (real) (nVertLevels, nVertices, Time)

Units	s ⁻¹
Description	absolute vertical vorticity at a vertex
Accessed in code	as pv_vertex from the diag

q2 (real) (nCells, Time)

Units	kg kg ⁻¹
-------	---------------------



[MPAS Atmosphere](#)

Menu

Description	2-meter specific humidity
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as q2 from the diag_physics

qbuoy (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	TKE production - buoyancy
Allocated by	bl_mynn_in
Accessed in code	as qbuoy from the diag_physics

qc_bl (real) (nVertLevels, nCells, Time)

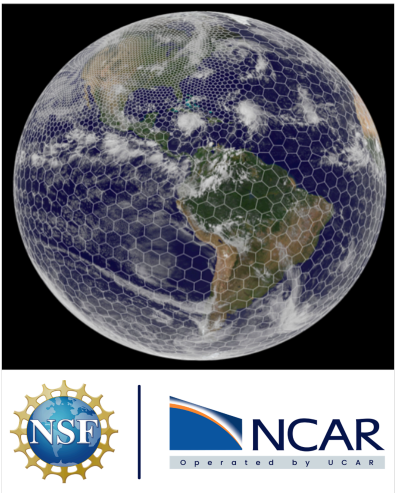
Units	kg kg^{-1}
Description	
Allocated by	bl_mynn_in
Accessed in code	as qc_bl from the diag_physics

qc_cu (real) (nVertLevels, nCells, Time)

Units	kg kg^{-1}
Description	in-cloud cloud water mixing ratio in Grell-Freitas cloud model
Allocated by	cu_grell_freitas_in
Accessed in code	as qc_cu from the diag_physics

qcg (real) (nCells, Time)

Units	kg kg^{-1}
Description	cloud water mixing ratio at the ground surface
Allocated by	bl_mynn_in
Accessed in code	as qcg from the diag_physics



[MPAS Atmosphere](#)

Menu

qdiss (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	TKE dissipation
Allocated by	bl_mynn_in
Accessed in code	as qdiss from the diag_physics

qfx (real) (nCells, Time)

Units	$\text{kg m}^{-2} \text{s}^{-1}$
Description	upward moisture flux at the surface
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as qfx from the diag_physics

qgh (real) (nCells, Time)

Units	kg kg^{-1}
Description	lowest level saturation mixing ratio
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as qgh from the diag_physics

qi_bl (real) (nVertLevels, nCells, Time)

Units	kg kg^{-1}
Description	
Allocated by	bl_mynn_in
Accessed in code	as qi_bl from the diag_physics

qi_cu (real) (nVertLevels, nCells, Time)

Units	kg kg^{-1}
Description	in-cloud cloud ice mixing ratio in Grell-Freitas cloud model
Allocated by	cu_grell_freitas_in
Accessed in code	as qi_cu from the diag_physics



[MPAS Atmosphere](#)

Menu

qke (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	twice turbulent kinetic energy
Allocated by	bl_mynn_in
Accessed in code	as qke from the diag_physics

qke_adv (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	twice turbulent kinetic energy
Allocated by	bl_mynn_in
Accessed in code	as qke_adv from the diag_physics

qsfc (real) (nCells, Time)

Units	kg kg^{-1}
Description	specific humidity at lower boundary
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as qsfc from the diag_physics

qshear (real) (nVertLevels, nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	TKE production - shear
Allocated by	bl_mynn_in
Accessed in code	as qshear from the diag_physics

qsq (real) (nVertLevels, nCells, Time)

Units	$\text{kg kg}^{-1 2}$
Description	liquid water variance



[MPAS Atmosphere](#)

Menu

Allocated by	bl_mynn_in
Accessed in code	as qsq from the diag_physics

qv_init (real) (nVertLevels)

Units	kg kg ⁻¹
Description	qv reference profile
Accessed in code	as qv_init from the mesh

qwt (real) (nVertLevels, nCells, Time)

Units	m ² s ⁻²
Description	TKE vertical distribution
Allocated by	bl_mynn_in
Accessed in code	as qwt from the diag_physics

rainc (real) (nCells, Time)

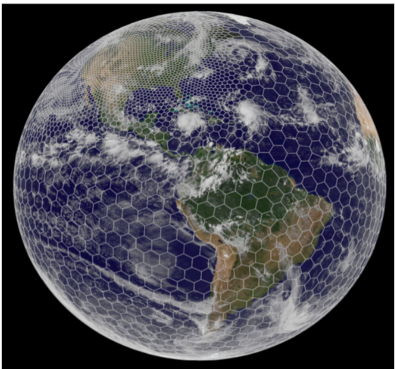
Units	mm
Description	accumulated convective precipitation
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as rainc from the diag_physics

raincv (real) (nCells, Time)

Units	mm
Description	time-step convective precipitation
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as raincv from the diag_physics

rainnc (real) (nCells, Time)

Units	mm
Description	accumulated total grid-scale precipitation



[MPAS Atmosphere](#)

Menu

Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in
Accessed in code	as rainnc from the diag_physics

rainncv (real) (nCells, Time)

Units	mm
Description	time-step total grid-scale precipitation
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in
Accessed in code	as rainncv from the diag_physics

rainprod (real) (nVertLevels, nCells, Time)

Units	s ⁻¹
Description	rain production rate
Allocated by	mp_thompson_in
Accessed in code	as rainprod from the diag_physics

rdzu (real) (nVertLevels)

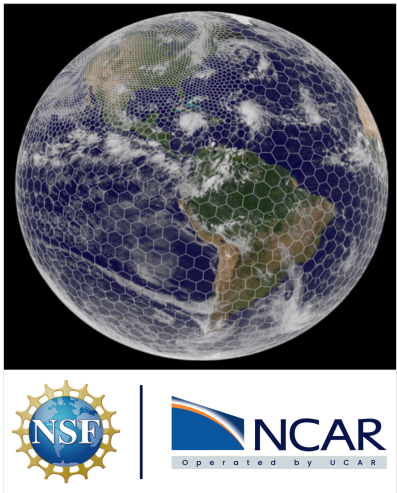
Units	unitless
Description	Reciprocal dzu
Accessed in code	as rdzu from the mesh

rdzw (real) (nVertLevels)

Units	unitless
Description	Reciprocal dzw
Accessed in code	as rdzw from the mesh

re_cloud (real) (nVertLevels, nCells, Time)

Units	m
Description	effective radius of cloud water droplets
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as re_cloud from the diag_physics



[MPAS Atmosphere](#)

Menu

re_ice (real) (nVertLevels, nCells, Time)

Units	m
Description	effective radius of cloud ice crystals
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as re_ice from the diag_physics

re_snow (real) (nVertLevels, nCells, Time)

Units	m
Description	effective radius of snow crystals
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as re_snow from the diag_physics

refl10cm_1km (real) (nCells, Time)

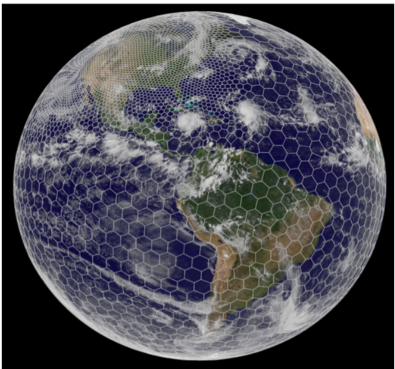
Units	dBZ
Description	diagnosed 10 cm radar reflectivity at 1 km AGL
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as refl10cm_1km from the diag_physics

refl10cm_1km_max (real) (nCells, Time)

Units	dBZ
Description	maximum diagnosed 10 cm radar reflectivity at 1 km AGL since last output time
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as refl10cm_1km_max from the diag_physics

refl10cm_max (real) (nCells, Time)

Units	dBZ
Description	10 cm maximum radar reflectivity
Allocated by	mp_thompson_in, mp_wsm6_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as refl10cm_max from the diag_physics
------------------	---------------------------------------

regime (real) (nCells, Time)

Units	unitless
Description	flag indicating the PBL regime (stable,unstable,...)
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as regime from the diag_physics

relhum (real) (nVertLevels, nCells, Time)

Units	percent
Description	Relative humidity
Accessed in code	as relhum from the diag

relhum_100hPa (real) (nCells, Time)

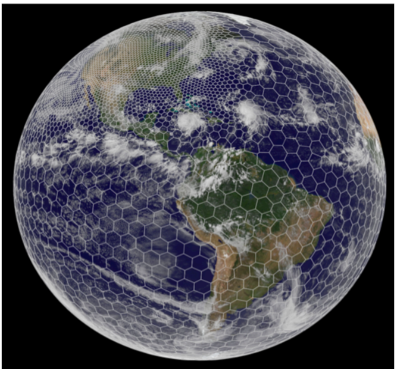
Units	percent
Description	Relative humidity vertically interpolated to 100 hPa
Accessed in code	as relhum_100hPa from the diag

relhum_200hPa (real) (nCells, Time)

Units	percent
Description	Relative humidity vertically interpolated to 200 hPa
Accessed in code	as relhum_200hPa from the diag

relhum_250hPa (real) (nCells, Time)

Units	percent
Description	Relative humidity vertically interpolated to 250 hPa



[MPAS Atmosphere](#)

Menu

Accessed in code	as relhum_250hPa from the diag
---------------------	--------------------------------------

relhum_500hPa (real) (nCells, Time)

Units	percent
Description	Relative humidity vertically interpolated to 500 hPa
Accessed in code	as relhum_500hPa from the diag

relhum_50hPa (real) (nCells, Time)

Units	percent
Description	Relative humidity vertically interpolated to 50 hPa
Accessed in code	as relhum_50hPa from the diag

relhum_700hPa (real) (nCells, Time)

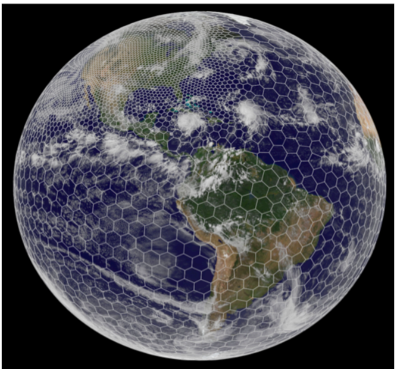
Units	percent
Description	Relative humidity vertically interpolated to 700 hPa
Accessed in code	as relhum_700hPa from the diag

relhum_850hPa (real) (nCells, Time)

Units	percent
Description	Relative humidity vertically interpolated to 850 hPa
Accessed in code	as relhum_850hPa from the diag

relhum_925hPa (real) (nCells, Time)

Units	percent
-------	---------



[MPAS Atmosphere](#)

Menu

Description	Relative humidity vertically interpolated to 925 hPa
Accessed in code	as relhum_925hPa from the diag

rho (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	Dry air density
Accessed in code	as rho from the diag

rho_amb (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	Dry air density increment
Accessed in code	as rho from the tend_iau

rho_base (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	Base state dry air density
Accessed in code	as rho_base from the diag

rho_edge (real) (nVertLevels, nEdges, Time)

Units	kg m ⁻³
Description	rho_zz averaged from cell centers to the cell edge
Accessed in code	as rho_edge from the diag

rho_p (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
-------	--------------------



[MPAS Atmosphere](#)

Menu

Description	rho/zz perturbation from the reference state value, advanced over acoustic steps
Accessed in code	as rho_p from the diag

rho_p_save (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	predicted value rho_p, saved before acoustic steps
Accessed in code	as rho_p_save from the diag

rho_pp (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	rho/zz perturbation from rho_pp, advanced over acoustic steps
Accessed in code	as rho_pp from the diag

rho_zz (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
Description	Dry air density divided by d(zeta)/dz
Accessed in code	as rho_zz from the state

rho_zz_old_split (real) (nVertLevels, nCells, Time)

Units	kg m ⁻³
-------	--------------------



[MPAS Atmosphere](#)

Menu

Description	rho/zz
Accessed in code	as rho_zz_old_split from the diag

rmol (real) (nCells, Time)

Units	unitless
Description	1./L Monin Obukhov length
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as rmol from the diag_physics

rniblten (real) (nVertLevels, nCells, Time)

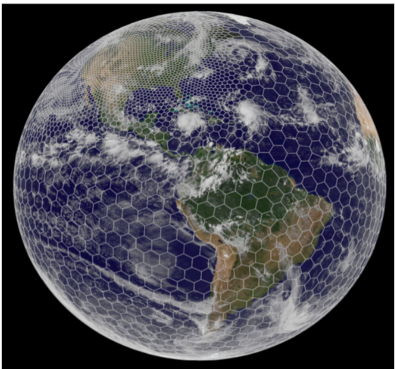
Units	nb kg ⁻¹ s ⁻¹
Description	tendency of cloud ice number concentration due to pbl processes
Allocated by	bl_mynn_in
Accessed in code	as rniblten from the tend_physics

rqcblten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of cloud water mixing ratio due to pbl processes
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as rqcblten from the tend_physics

rqccuten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of cloud water mixing ratio due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as rqqcuten from the tend_physics
------------------	-----------------------------------

rqibltten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of cloud ice mixing ratio due to pbl processes
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as rqibltten from the tend_physics

rqicuten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of cloud ice mixing ratio due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as rqicuten from the tend_physics

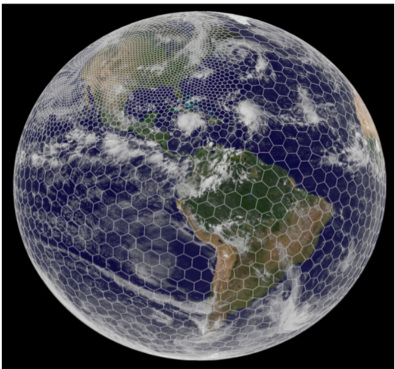
rqrcuten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of rain mixing ratio due to cumulus convection
Allocated by	cu_kain_fritsch_in
Accessed in code	as rqrcuten from the tend_physics

rqsbltten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of snow mixing ratio due to pbl processes
Allocated by	bl_mynn_in
Accessed in code	as rqsbltten from the tend_physics

rqscuten (real) (nVertLevels, nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of snow mixing ratio due to cumulus convection
Allocated by	cu_kain_fritsch_in
Accessed in code	as rqscuten from the tend_physics

rqvblten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of water vapor mixing ratio due to pbl processes
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as rqvblten from the tend_physics

rqvcuten (real) (nVertLevels, nCells, Time)

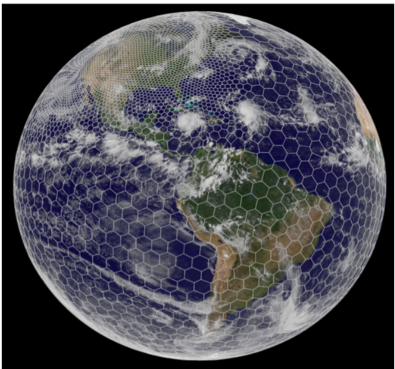
Units	kg kg ⁻¹ s ⁻¹
Description	tendency of water vapor mixing ratio due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as rqvcuten from the tend_physics

rqvdynten (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	tendency of water vapor due to horizontal and vertical advections
Allocated by	cu_grell_freitas_in, cu_ntiedtke_in
Accessed in code	as rqvdynten from the tend_physics

rre_cloud (real) (nVertLevels, nCells, Time)

Units	microns
-------	---------



[MPAS Atmosphere](#)

Menu

Description	effective radius of cloud water droplets calculated in RRTMG radiation
Accessed in code	as <code>rre_cloud</code> from the <code>diag_physics</code>

rre_ice (real) (nVertLevels, nCells, Time)

Units	microns
Description	effective radius of cloud ice crystals calculated in RRTMG radiation
Accessed in code	as <code>rre_ice</code> from the <code>diag_physics</code>

rre_snow (real) (nVertLevels, nCells, Time)

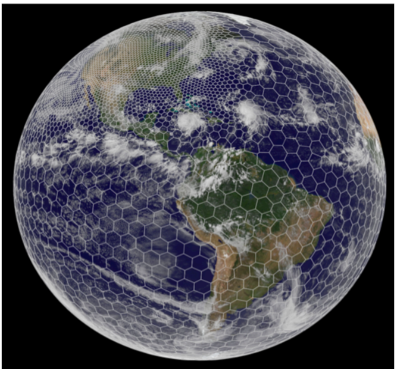
Units	microns
Description	effective radius of snow crystals calculated in RRTMG radiation
Accessed in code	as <code>rre_snow</code> from the <code>diag_physics</code>

rt_diabatic_tend (real) (nVertLevels, nCells, Time)

Units	K s^{-1}
Description	Tendency of modified potential temperature due to cloud microphysics
Accessed in code	as <code>rt_diabatic_tend</code> from the <code>tend</code>

rthblten (real) (nVertLevels, nCells, Time)

Units	K s^{-1}
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	tendency of potential temperature due to pbl processes
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as rthblten from the tend_physics

rthcuten (real) (nVertLevels, nCells, Time)

Units	K s^{-1}
Description	tendency of potential temperature due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_kain_fritsch_in, cu_ntiedtke_in
Accessed in code	as rthcuten from the tend_physics

rthdynten (real) (nVertLevels, nCells, Time)

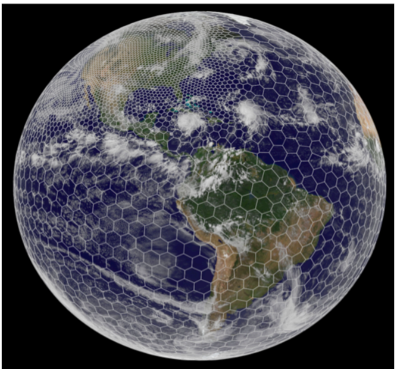
Units	K s^{-1}
Description	tendency of temperature due to horizontal and vertical advections
Accessed in code	as rthdynten from the tend_physics

rtheta_base (real) (nVertLevels, nCells, Time)

Units	kg K m^{-3}
Description	reference state $\rho \cdot \theta / z$
Accessed in code	as rtheta_base from the diag

rtheta_p (real) (nVertLevels, nCells, Time)

Units	kg K m^{-3}
Description	$\rho \cdot \theta_m / z$ perturbation from the reference state value



[MPAS Atmosphere](#)

Menu

Accessed in code	as rtheta_p from the diag
---------------------	------------------------------

rtheta_p_save (real) (nVertLevels, nCells, Time)

Units	kg K m ⁻³
Description	predicted value rtheta_p, saved before acoustic steps
Accessed in code	as rtheta_p_save from the diag

rtheta_pp (real) (nVertLevels, nCells, Time)

Units	kg K m ⁻³
Description	rho*theta_m/zz perturbation from rtheta_p
Accessed in code	as rtheta_pp from the diag

rtheta_pp_old (real) (nVertLevels, nCells, Time)

Units	kg K m ⁻³
Description	old time level values of rho*theta_m/zz perturbation from rtheta_p, used in 3D divergence damping
Accessed in code	as rtheta_pp_old from the diag

rthratenlw (real) (nVertLevels, nCells, Time)

Units	K s ⁻¹
Description	tendency of potential temperature due to long wave radiation
Accessed in code	as rthratenlw from the tend_physics

rthratensw (real) (nVertLevels, nCells, Time)

Units	K s ⁻¹
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	tendency of potential temperature due to short wave radiation
Accessed in code	as rthratensw from the tend_physics

ru (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	horizontal momentum at cell edge ($\rho \cdot u / \Delta z$)
Accessed in code	as ru from the diag

ru_p (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	acoustic perturbation horizontal momentum at cell edge ($\rho \cdot u / \Delta z$)
Accessed in code	as ru_p from the diag

ru_save (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	predicted value of horizontal momentum, saved before acoustic steps
Accessed in code	as ru_save from the diag

ruAvg (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	time-averaged $\rho \cdot u / \Delta z$ used in scalar transport



[MPAS Atmosphere](#)

Menu

Accessed in code	as ruAvg from the diag
---------------------	------------------------------

ruAvg_split (real) (nVertLevels, nEdges, Time)

Units	$\text{kg m}^{-2} \text{s}^{-1}$
Description	time-averaged $\rho u / \Delta z$ used in scalar transport
Accessed in code	as ruAvg_split from the diag

rubldiff (real) (nVertLevels, nCells, Time)

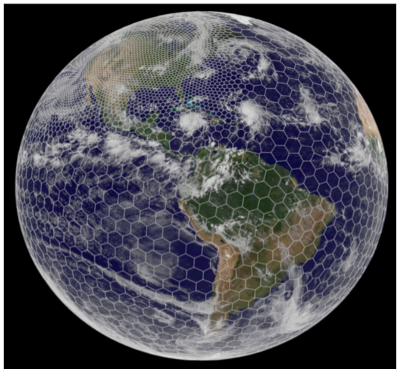
Units	m s^{-2}
Description	change in PBL zonal wind tendency due to gravity wave drag over orography
Accessed in code	as rubldiff from the diag_physics

rublten (real) (nVertLevels, nCells, Time)

Units	$\text{m s}^{-1} \text{s}^{-1}$
Description	tendency of zonal wind due to pbl processes
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as rublten from the tend_physics

rublten_Edge (real) (nVertLevels, nEdges, Time)

Units	
Description	
Accessed in code	as rublten_Edge from the tend_physics



[MPAS Atmosphere](#)

Menu

rucuten (real) (nVertLevels, nCells, Time)

Units	$\text{m s}^{-1} \text{ s}^{-1}$
Description	tendency of zonal wind due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_ntiedtke_in
Accessed in code	as rucuten from the tend_physics

rucuten_Edge (real) (nVertLevels, nEdges, Time)

Units	
Description	
Accessed in code	as rucuten_Edge from the tend_physics

rv (real) (nVertLevels, nEdges, Time)

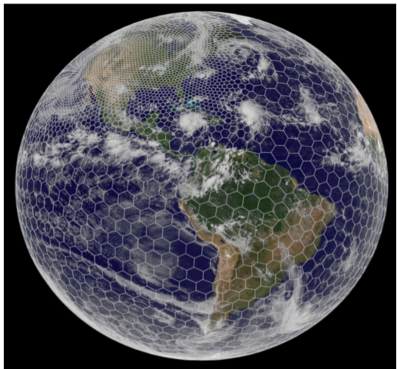
Units	???
Description	???
Accessed in code	as rv from the diag

rvbldiff (real) (nVertLevels, nCells, Time)

Units	m s^{-2}
Description	change in PBL meridional wind tendency due to gravity wave drag over orography
Accessed in code	as rvbldiff from the diag_physics

rvblten (real) (nVertLevels, nCells, Time)

Units	$\text{m s}^{-1} \text{ s}^{-1}$
Description	tendency of meridional wind due to pbl processes
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as rvblten from the tend_physics



rvcuten (real) (nVertLevels, nCells, Time)

Units	$\text{m s}^{-1} \text{ s}^{-1}$
Description	tendency of meridional wind due to cumulus convection
Allocated by	cu_grell_freitas_in, cu_ntiedtke_in
Accessed in code	as rvcuten from the tend_physics

rw (real) (nVertLevelsP1, nCells, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	$\rho \cdot \omega / \text{zz}$ carried at w points
Accessed in code	as rw from the diag

rw_p (real) (nVertLevelsP1, nCells, Time)

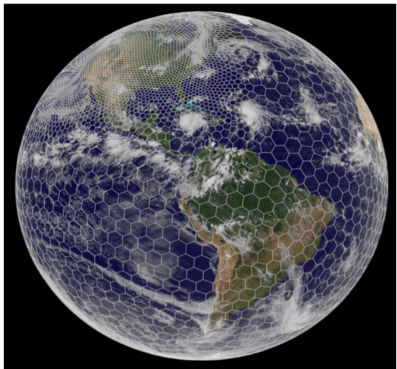
Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	acoustic perturbation $\rho \cdot \omega / \text{zz}$ carried at w points
Accessed in code	as rw_p from the diag

rw_save (real) (nVertLevelsP1, nCells, Time)

Units	$\text{kg m}^{-2} \text{ s}^{-1}$
Description	predicted value of $\rho \cdot \omega / \text{zz}$, saved before acoustic steps
Accessed in code	as rw_save from the diag

scalars (real) (nVertLevels, nCells, Time)

Description	scalars is a variable array used in code which is made up of the constituent fields listed below.
-------------	---



[MPAS Atmosphere](#)

Menu

Accessed in code	scalars from the state pool
------------------	-----------------------------

qv – constituent field of var_array scalars

Description	Water vapor mixing ratio
Units	kg kg ⁻¹
Array Group	moist

qc – constituent field of var_array scalars

Description	Cloud water mixing ratio
Units	kg kg ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, bl_ysu_in, cu_ntiedtke_in, mp_kessler_in, mp_thompson_in, mp_wsm6_in

qr – constituent field of var_array scalars

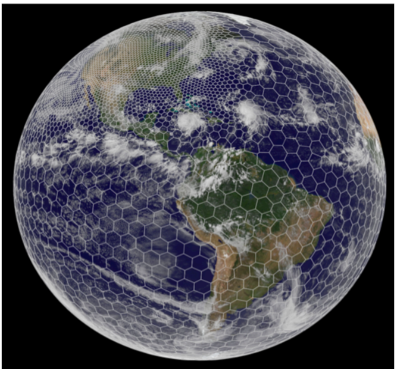
Description	Rain water mixing ratio
Units	kg kg ⁻¹
Array Group	moist
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in

qi – constituent field of var_array scalars

Description	Ice mixing ratio
Units	kg kg ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, bl_ysu_in, cu_ntiedtke_in, mp_thompson_in, mp_wsm6_in

qs – constituent field of var_array scalars

Description	Snow mixing ratio
Units	kg kg ⁻¹
Array Group	moist



[MPAS Atmosphere](#)

Menu

Allocated by	bl_mynn_in, mp_thompson_in, mp_wsm6_in
--------------	--

qg – constituent field of var_array scalars

Description	Graupel mixing ratio
Units	kg kg ⁻¹
Array Group	moist
Allocated by	mp_thompson_in, mp_wsm6_in

ni – constituent field of var_array scalars

Description	Cloud ice number concentration
Units	nb kg ⁻¹
Array Group	number
Allocated by	bl_mynn_in, mp_thompson_in

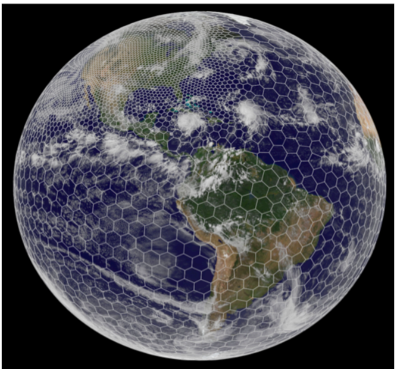
nr – constituent field of var_array scalars

Description	Rain number concentration
Units	nb kg ⁻¹
Array Group	number
Allocated by	mp_thompson_in

scalars_amb (real) (nVertLevels, nCells, Time)

Description	scalars_amb is a variable array used in code which is made up of the constituent fields listed below.
Accessed in code	scalars from the tend_iau pool

qv_amb – constituent field of var_array scalars_amb



[MPAS Atmosphere](#)

Menu

Description	Water vapor mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist

qc_amb – constituent field of var_array scalars_amb

Description	Cloud water mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, bl_yсу_in, cu_ntiedtke_in, mp_kessler_in, mp_thompson_in, mp_wsm6_in

qr_amb – constituent field of var_array scalars_amb

Description	Rain water mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in

qi_amb – constituent field of var_array scalars_amb

Description	Ice mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, bl_yсу_in, cu_ntiedtke_in, mp_thompson_in, mp_wsm6_in

qs_amb – constituent field of var_array scalars_amb

Description	Snow mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist



[MPAS Atmosphere](#)

Menu

Allocated by	mp_thompson_in, mp_wsm6_in
--------------	----------------------------

qg_amb – constituent field of var_array scalars_amb

Description	Graupel mixing ratio increment
Units	kg kg ⁻¹
Array Group	moist
Allocated by	mp_thompson_in, mp_wsm6_in

scalars_tend (real) (nVertLevels, nCells, Time)

Description	scalars_tend is a variable array used in code which is made up of the constituent fields listed below.
Accessed in code	scalars_tend from the tend pool

tend_qv – constituent field of var_array scalars_tend

Description	Tendency of water vapor mass per unit volume divided by d(zeta)/dz
Units	kg m ⁻³ s ⁻¹
Array Group	moist

tend_qc – constituent field of var_array scalars_tend

Description	Tendency of cloud water mass per unit volume divided by d(zeta)/dz
Units	kg m ⁻³ s ⁻¹
Array Group	moist



[MPAS Atmosphere](#)

Menu

Allocated by	bl_mynn_in, bl_ysu_in, cu_ntiedtke_in, mp_kessler_in, mp_thompson_in, mp_wsm6_in
--------------	---

tend_qr – constituent field of var_array
scalars_tend

Description	Tendency of rain water mass per unit volume divided by d(zeta)/dz
Units	kg m ⁻³ s ⁻¹
Array Group	moist
Allocated by	mp_kessler_in, mp_thompson_in, mp_wsm6_in

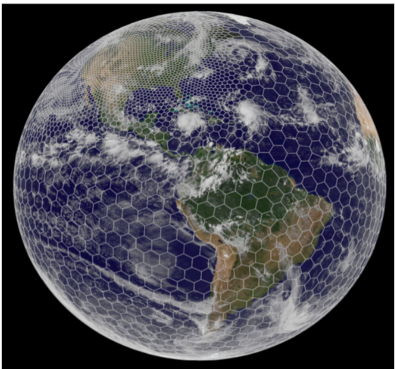
tend_qi – constituent field of var_array
scalars_tend

Description	Tendency of ice mass per unit volume divided by d(zeta)/dz
Units	kg m ⁻³ s ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, bl_ysu_in, cu_ntiedtke_in, mp_thompson_in, mp_wsm6_in

tend_qs – constituent field of var_array
scalars_tend

Description	Tendency of snow mass per unit volume divided by d(zeta)/dz
Units	kg m ⁻³ s ⁻¹
Array Group	moist
Allocated by	bl_mynn_in, mp_thompson_in, mp_wsm6_in

tend_qg – constituent field of var_array
scalars_tend



[MPAS Atmosphere](#)

Menu

Description	Tendency of graupel mass per unit volume divided by $d(\text{zeta})/dz$
Units	$\text{kg m}^{-3} \text{s}^{-1}$
Array Group	moist
Allocated by	mp_thompson_in, mp_wsm6_in

tend_ni – constituent field of var_array scalars_tend

Description	Tendency of cloud ice number concentration multiplied by dry air density divided by $d(\text{zeta})/dz$
Units	$\text{nb m}^{-3} \text{s}^{-1}$
Array Group	number
Allocated by	bl_mynn_in, mp_thompson_in

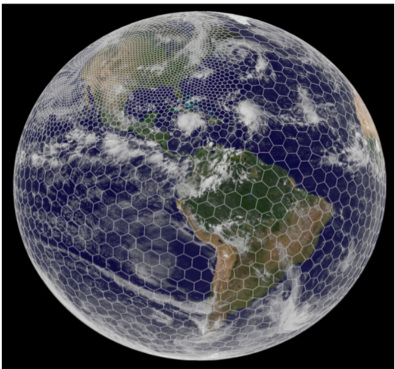
tend_nr – constituent field of var_array scalars_tend

Description	Tendency of rain number concentration multiplied by dry air density divided by $d(\text{zeta})/dz$
Units	$\text{nb m}^{-3} \text{s}^{-1}$
Array Group	number
Allocated by	mp_thompson_in

scale (real) (nVertLevels, TWO, nCells)

Units	•
Description	Scaling coefficients for monotonic-limited horizontal fluxes
Accessed in code	as scale from the halo_scratch

seaice (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	unitless
Description	sea-ice flag (0=no seaice; =1 otherwise)
Accessed in code	as seaice from the sfc_input

sfc_albbck (real) (nCells, Time)

Units	unitless
Description	background surface albedo
Accessed in code	as sfc_albbck from the sfc_input

sfc_albedo (real) (nCells, Time)

Units	unitless
Description	surface albedo
Accessed in code	as sfc_albedo from the diag_physics

sfc_emibck (real) (nCells, Time)

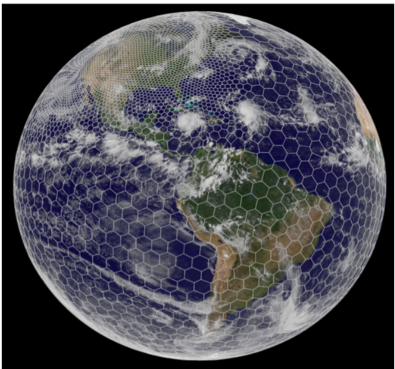
Units	unitless
Description	background surface emissivity
Accessed in code	as sfc_emibck from the diag_physics

sfc_emiss (real) (nCells, Time)

Units	unitless
Description	surface emissivity
Accessed in code	as sfc_emiss from the diag_physics

sfcrunoff (real) (nCells, Time)

Units	mm
Description	surface runoff



[MPAS Atmosphere](#)

Menu

Accessed in code	as sfcrunoff from the diag_physics
---------------------	--

sh2o (real) (nSoilLevels, nCells, Time)

Units	m3 m ⁻³
Description	soil equivalent liquid water
Accessed in code	as sh2o from the sfc_input

sh3d (real) (nVertLevels, nCells, Time)

Units	unitless
Description	stability function for heat
Allocated by	bl_mynn_in
Accessed in code	as sh3d from the diag_physics

shdmax (real) (nCells)

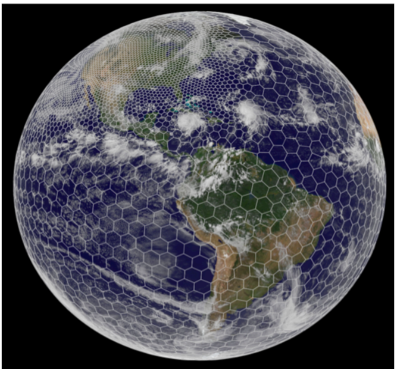
Units	unitless
Description	maximum fractional coverage of annual green vegetation fraction
Accessed in code	as shdmax from the sfc_input

shdmin (real) (nCells)

Units	unitless
Description	minimum fractional coverage of annual green vegetation fraction
Accessed in code	as shdmin from the sfc_input

skintemp (real) (nCells, Time)

Units	K
-------	---



[MPAS Atmosphere](#)

Menu

Description	ground or water surface temperature
Accessed in code	as skintemp from the sfc_input

sm3d (real) (nVertLevels, nCells, Time)

Units	unitless
Description	stability function for moisture
Allocated by	bl_mynn_in
Accessed in code	as sm3d from the diag_physics

smcrel (real) (nSoilLevels, nCells, Time)

Units	m3 m ⁻³
Description	soil moisture threshold below which transpiration begins to stress
Accessed in code	as smcrel from the sfc_input

smois (real) (nSoilLevels, nCells, Time)

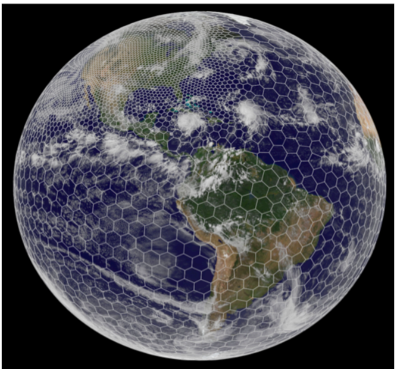
Units	m3 m ⁻³
Description	soil moisture
Accessed in code	as smois from the sfc_input

smstav (real) (nCells, Time)

Units	unitless
Description	surface moisture availability
Accessed in code	as smstav from the diag_physics

smstot (real) (nCells, Time)

Units	m ³ m ^{sup:-3`}
Description	total soil mositure



[MPAS Atmosphere](#)

Menu

Accessed in code	as smstot from the diag_physics
---------------------	---------------------------------------

snoalb (real) (nCells)

Units	unitless
Description	annual maximum snow albedo
Accessed in code	as snoalb from the sfc_input

snopcx (real) (nCells, Time)

Units	W m ⁻²
Description	snow phase change heat flux
Accessed in code	as snopcx from the diag_physics

snotime (real) (nCells, Time)

Units	unitless
Description	initial number of time-steps since last snow fall
Accessed in code	as snotime from the diag_physics

snow (real) (nCells, Time)

Units	kg m ⁻²
Description	snow water equivalent
Accessed in code	as snow from the sfc_input

snowc (real) (nCells, Time)

Units	unitless
Description	flag for snow on ground (=0 no snow; =1,otherwise)
Accessed in code	as snowc from the sfc_input



[MPAS Atmosphere](#)

Menu

snowh (real) (nCells, Time)

Units	m
Description	physical snow depth
Accessed in code	as snowh from the sfc_input

snownc (real) (nCells, Time)

Units	mm
Description	accumulated grid-scale precipitation of snow
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as snownc from the diag_physics

snowncv (real) (nCells, Time)

Units	mm
Description	time-step grid-scale precipitation of snow
Allocated by	mp_thompson_in, mp_wsm6_in
Accessed in code	as snowncv from the diag_physics

specchum (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹
Description	Specific humidity
Allocated by	jedi_da
Accessed in code	as specchum from the diag

specZoneMaskCell (real) (nCells)

Units	•
Description	0/1 mask on cells, defined as 1 for cells in the limited-area specified zone
Accessed in code	as specZoneMaskCell from the mesh

specZoneMaskEdge (real) (nEdges)

Units	•
-------	---



[MPAS Atmosphere](#)

Menu

Description	0/1 mask on edges, defined as 1 for edges in the limited-area specified zone
Accessed in code	as specZoneMaskEdge from the mesh

specZoneMaskVertex (real) (nVertices)

Units	•
Description	0/1 mask on vertices, defined as 1 for vertices in the limited-area specified zone
Accessed in code	as specZoneMaskVertex from the mesh

sr (real) (nCells, Time)

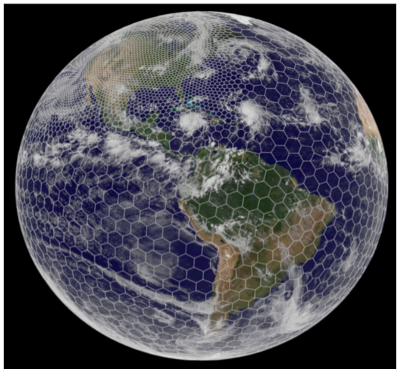
Units	unitless
Description	time-step ratio of frozen versus total grid-scale precipitation
Accessed in code	as sr from the diag_physics

srh_0_1km (real) (nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	Storm relative helicity, 0-1` km AGL
Accessed in code	as srh_0_1km from the diag

srh_0_3km (real) (nCells, Time)

Units	$\text{m}^2 \text{s}^{-2}$
Description	Storm relative helicity, 0-3` km AGL
Accessed in code	as srh_0_3km from the diag



[MPAS Atmosphere](#)

Menu

sst (real) (nCells, Time)

Units	K
Description	sea-surface temperature
Accessed in code	as sst from the sfc_input

sstsk (real) (nCells, Time)

Units	K
Description	skin sea-surface temperature
Accessed in code	as sstsk from the diag_physics

sstsk_dtc (real) (nCells, Time)

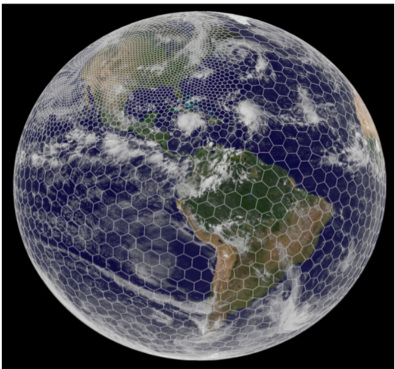
Units	K
Description	skin sea-surface temperature cooling
Accessed in code	as sstsk_dtc from the diag_physics

sstsk_dtw (real) (nCells, Time)

Units	K
Description	skin sea-surface temperature warming
Accessed in code	as sstsk_dtw from the diag_physics

sub_qv (real) (nVertLevels, nCells, Time)

Units	kg kg ⁻¹ s ⁻¹
Description	EDMF subsidence of water vapor mixing ratio in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as sub_qv from the diag_physics



[MPAS Atmosphere](#)

Menu

sub_thl (real) (nVertLevels, nCells, Time)

Units	K s^{-1}
Description	EDMF subsidence of liquid-ice potential temperature in the MYNN PBL scheme
Allocated by	bl_mynn_in
Accessed in code	as sub_thl from the diag_physics

surface_pressure (real) (nCells, Time)

Units	Pa
Description	Diagnosed surface pressure
Accessed in code	as surface_pressure from the diag

swcf (real) (nCells, Time)

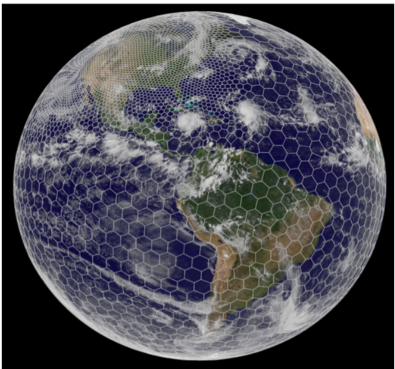
Units	W m^{-2}
Description	top-of-atmosphere cloud shortwave radiative forcing
Accessed in code	as swcf from the diag_physics

swddif (real) (nCells, Time)

Units	W m^{-2}
Description	shortwave surface downward diffuse irradiance
Accessed in code	as swddif from the diag_physics

swddir (real) (nCells, Time)

Units	W m^{-2}
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	shortwave surface downward direct irradiance
Accessed in code	as swddir from the diag_physics

swddni (real) (nCells, Time)

Units	W m ⁻²
Description	shortwave surface downward direct normal irradiance
Accessed in code	as swddni from the diag_physics

swdnb (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky downward surface shortwave radiation flux
Accessed in code	as swdnb from the diag_physics

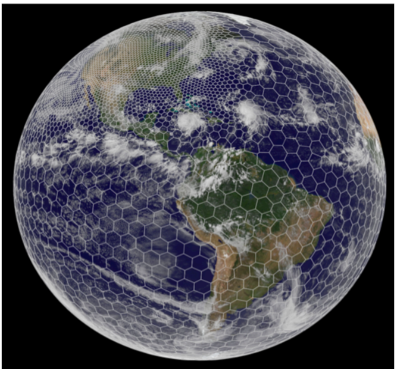
swdnbc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky downward surface shortwave radiation flux
Accessed in code	as swdnbc from the diag_physics

swdnflx (real) (nVertLevelsP2, nCells, Time)

Units	•
Description	•
Accessed in code	as swdnflx from the diag_physics

swdnflxc (real) (nVertLevelsP2, nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	
Description	
Accessed in code	as swdnflxc from the diag_physics

swdnt (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky downward top-of-atmosphere shortwave radiation flux
Accessed in code	as swdnt from the diag_physics

swdntc (real) (nCells, Time)

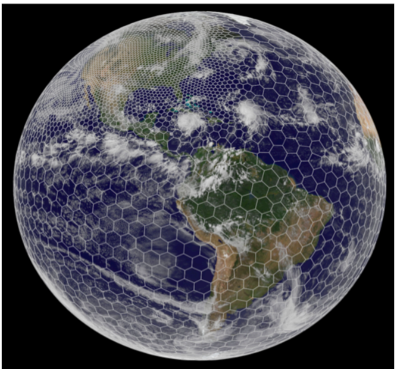
Units	W m ⁻²
Description	clear-sky downward top-of-atmosphere shortwave radiation flux
Accessed in code	as swdntc from the diag_physics

swupb (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky upward surface shortwave radiation flux
Accessed in code	as swupb from the diag_physics

swupbc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky upward surface shortwave radiation flux



[MPAS Atmosphere](#)

Menu

Accessed in code	as swupbc from the diag_physics
---------------------	---------------------------------------

swupflx (real) (nVertLevelsP2, nCells, Time)

Units	•
Description	•
Accessed in code	as swupflx from the diag_physics

swupflxc (real) (nVertLevelsP2, nCells, Time)

Units	•
Description	•
Accessed in code	as swupflxc from the diag_physics

swupt (real) (nCells, Time)

Units	W m ⁻²
Description	all-sky upward top- of- atmosphere shortwave radiation flux
Accessed in code	as swupt from the diag_physics

swuptc (real) (nCells, Time)

Units	W m ⁻²
Description	clear-sky upward top- of- atmosphere shortwave radiation flux
Accessed in code	as swuptc from the diag_physics

t2m (real) (nCells, Time)

Units	K
Description	2-meter temperature
Allocated by	bl_mynn_in, bl_ysu_in



[MPAS Atmosphere](#)

Menu

Accessed in code	as t2m from the diag_physics
---------------------	------------------------------------

t_init (real) (nVertLevels, nCells)

Units	K
Description	theta reference profile
Accessed in code	as t_init from the mesh

t_iso_levels (real) (nIsoLevelsT)

Units	Pa
Description	Levels for vertical interpolation of temperature to isobaric surfaces
Accessed in code	as t_iso_levels from the diag

t_isobaric (real) (nIsoLevelsT, nCells, Time)

Units	K
Description	Temperature interpolated to isobaric surfaces defined in t_iso_levels
Accessed in code	as t_isobaric from the diag

t_oml (real) (nCells, Time)

Units	K
Description	ocean mixed layer temperature
Accessed in code	as t_oml from the diag_physics

t_oml_200m_initial (real) (nCells, Time)

Units	K
-------	---



[MPAS Atmosphere](#)

Menu

Description	ocean mixed layer 200 m mean temperature at initial time
Accessed in code	as t_oml_200m_initial from the diag_physics

t_oml_initial (real) (nCells, Time)

Units	K
Description	ocean mixed layer temperature at initial time
Accessed in code	as t_oml_initial from the diag_physics

tday_accum (real) (nCells, Time)

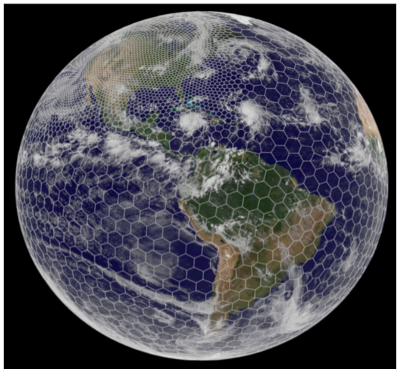
Units	K
Description	accumulated daily surface temperature for current day
Accessed in code	as tday_accum from the diag_physics

temperature (real) (nVertLevels, nCells, Time)

Units	K
Description	temperature
Allocated by	jedi_da
Accessed in code	as temperature from the diag

temperature_100hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 100 hPa
Accessed in code	as temperature_100hPa from the diag



[MPAS Atmosphere](#)

Menu

temperature_200hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 200 hPa
Accessed in code	as temperature_200hPa from the diag

temperature_250hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 250 hPa
Accessed in code	as temperature_250hPa from the diag

temperature_500hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 500 hPa
Accessed in code	as temperature_500hPa from the diag

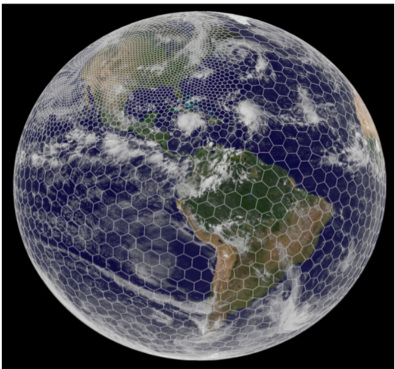
temperature_50hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 50 hPa
Accessed in code	as temperature_50hPa from the diag

temperature_700hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 700 hPa
Accessed in code	as temperature_700hPa from the diag

temperature_850hPa (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	K
Description	Temperature vertically interpolated to 850 hPa
Accessed in code	as temperature_850hPa from the diag

temperature_925hPa (real) (nCells, Time)

Units	K
Description	Temperature vertically interpolated to 925 hPa
Accessed in code	as temperature_925hPa from the diag

temperature_surface (real) (nCells, Time)

Units	K
Description	Temperature at midpoint of lowest model layer
Accessed in code	as temperature_surface from the diag

tend_rho (real) (nVertLevels, nCells, Time)

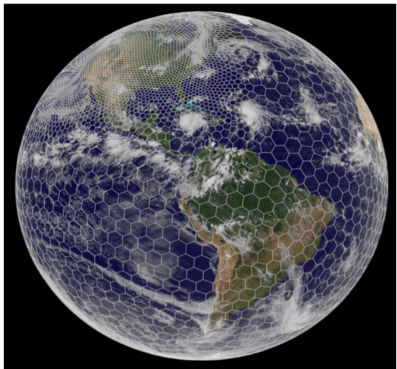
Units	$\text{kg m}^{-3} \text{ s}^{-1}$
Description	Tendency of dry density from dynamics
Accessed in code	as rho_zz from the tend

tend_rho_physics (real) (nVertLevels, nCells, Time)

Units	$\text{kg m}^{-3} \text{ s}^{-1}$
Description	Total dry air tendency from physics
Accessed in code	as tend_rho_physics from the tend_physics

tend_rtheta_physics (real) (nVertLevels, nCells, Time)

Units	kg K m^{-3}
-------	----------------------



[MPAS Atmosphere](#)

Menu

Description	Total rho*theta/zz tendency from physics
Accessed in code	as tend_rtheta_physics from the tend_physics

tend_ru_physics (real) (nVertLevels, nEdges, Time)

Units	kg m ⁻² s ⁻¹
Description	Total horizontal momentum tendency from physics
Accessed in code	as tend_ru_physics from the tend_physics

tend_sfc_pressure (real) (nCells, Time)

Units	Pa s ⁻¹
Description	Tendency of surface pressure
Accessed in code	as tend_sfc_pressure from the tend

tend_theta (real) (nVertLevels, nCells, Time)

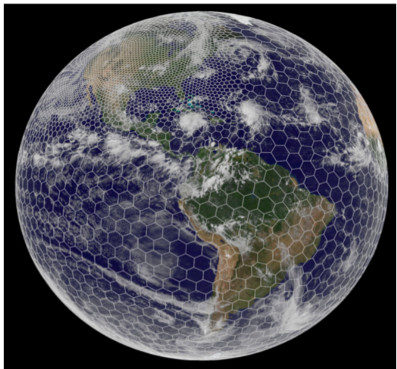
Units	kg K m ⁻³ s ⁻¹
Description	tendency of coupled potential temperature rho*theta_m/zz from dynamics and physics, updated each RK step
Accessed in code	as theta_m from the tend

tend_u (real) (nVertLevels, nEdges, Time)

Units	m s ⁻²
Description	Tendency of u from dynamics
Accessed in code	as u from the tend

tend_u_phys (real) (nVertLevels, nEdges, Time)

Units	m s ⁻²
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	Normal wind tendencies from physics parameterizations
Accessed in code	as tend_u_phys from the diag

tend_umerid (real) (nVertLevels, nCells, Time)

Units	$\text{m s}^{-1} \text{s}^{-1}$
Description	Total cell-centered meridional wind tendency from physics
Accessed in code	as tend_umerid from the tend_physics

tend_uzonal (real) (nVertLevels, nCells, Time)

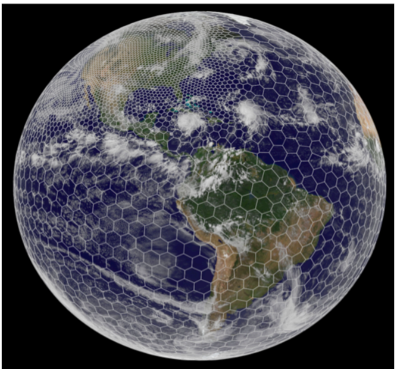
Units	$\text{m s}^{-1} \text{s}^{-1}$
Description	Total cell-centered zonal wind tendency from physics
Accessed in code	as tend_uzonal from the tend_physics

tend_w (real) (nVertLevelsP1, nCells, Time)

Units	m s^{-2}
Description	Tendency of w from dynamics
Accessed in code	as w from the tend

tend_w_buoy (real) (nVertLevelsP1, nCells, Time)

Units	m s^{-2}
Description	Tendency of w due to buoyancy force
Accessed in code	as w_buoy from the tend



[MPAS Atmosphere](#)

Menu

tend_w_pgf (real) (nVertLevelsP1, nCells, Time)

Units	m s ⁻²
Description	Tendency of w due to pressure gradient force
Accessed in code	as w_pgf from the tend

ter (real) (nCells)

Units	m
Description	terrain height
Accessed in code	as ter from the sfc_input

th2m (real) (nCells, Time)

Units	K
Description	2-meter potential temperature
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as th2m from the diag_physics

thc (real) (nCells, Time)

Units	Cal cm ⁻² K ⁻¹ s ^{-0.5}
Description	thermal inertia
Accessed in code	as thc from the diag_physics

theta (real) (nVertLevels, nCells, Time)

Units	K
Description	Potential temperature
Accessed in code	as theta from the diag

theta_amb (real) (nVertLevels, nCells, Time)

Units	K
-------	---



[MPAS Atmosphere](#)

Menu

Description	Potential temperature increment
Accessed in code	as theta from the tend_iau

theta_base (real) (nVertLevels, nCells, Time)

Units	K
Description	Base state potential temperature
Accessed in code	as theta_base from the diag

theta_m (real) (nVertLevels, nCells, Time)

Units	K
Description	Moist potential temperature: theta* (1+q_v*R_v/R_d)
Accessed in code	as theta_m from the state

theta_pv (real) (nCells, Time)

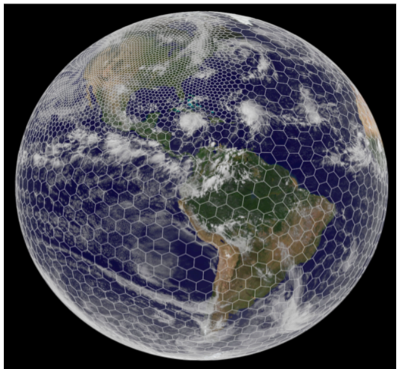
Units	K
Description	Potential temperature on dynamic tropopause
Accessed in code	as theta_pv from the diag

Time (real) (Time)

Units	seconds since YYYY-MM-DD hh:mm:ss
Description	CF-compliant valid time
Accessed in code	as Time from the state

tke_pbl (real) (nVertLevels, nCells, Time)

Units	m ² s ⁻²
-------	--------------------------------



[MPAS Atmosphere](#)

Menu

Description	turbulent kinetic energy from PBL
Allocated by	bl_mynn_in
Accessed in code	as tke_pbl from the diag_physics

tlag (real) (nLags, nCells, Time)

Units	K
Description	daily mean surface temperature of prior days
Accessed in code	as tlag from the diag_physics

tmn (real) (nCells, Time)

Units	K
Description	deep soil temperature
Accessed in code	as tmn from the sfc_input

tslb (real) (nSoilLevels, nCells, Time)

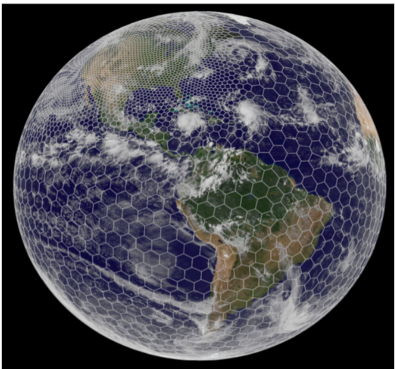
Units	K
Description	soil layer temperature
Accessed in code	as tslb from the sfc_input

tsq (real) (nVertLevels, nCells, Time)

Units	K ²
Description	liquid water potential temperature variance
Allocated by	bl_mynn_in
Accessed in code	as tsq from the diag_physics

tyear_accum (real) (nCells, Time)

Units	K
-------	---



[MPAS Atmosphere](#)

Menu

Description	accumulated yearly surface temperature for current year
Accessed in code	as tyear_accum from the diag_physics

tyear_mean (real) (nCells, Time)

Units	K
Description	annual mean surface temperature
Accessed in code	as tyear_mean from the diag_physics

u (real) (nVertLevels, nEdges, Time)

Units	m s ⁻¹
Description	Horizontal normal velocity at edges
Accessed in code	as u from the state

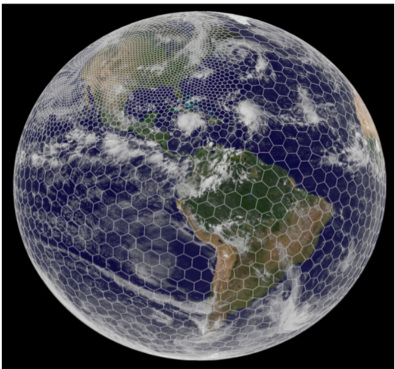
u10 (real) (nCells, Time)

Units	m s ⁻¹
Description	10-meter zonal wind
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as u10 from the diag_physics

u_amb (real) (nVertLevels, nEdges, Time)

Units	m s ⁻¹
Description	Horizontal normal velocity increment
Accessed in code	as u from the tend_iau

u_init (real) (nVertLevels)



[MPAS Atmosphere](#)

Menu

Units	m s^{-1}
Description	u reference profile
Accessed in code	as u_init from the mesh

u_pv (real) (nCells, Time)

Units	m s^{-1}
Description	Zonal wind on dynamic tropopause
Accessed in code	as u_pv from the diag

udrunoff (real) (nCells, Time)

Units	mm
Description	underground runoff
Accessed in code	as udrunoff from the diag_physics

umeridional_100hPa (real) (nCells, Time)

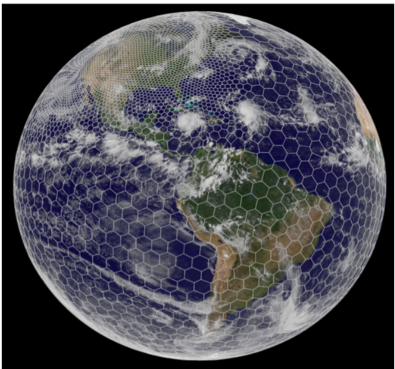
Units	m s^{-1}
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 100 hPa
Accessed in code	as umeridional_100hPa from the diag

umeridional_1km (real) (nCells, Time)

Units	m s^{-1}
Description	Meridional wind component at 1 km AGL
Accessed in code	as umeridional_1km from the diag

umeridional_200hPa (real) (nCells, Time)

Units	m s^{-1}
-------	-------------------



[MPAS](#)
[Atmosphere](#)

Menu

Description	Reconstructed meridional wind at cell centers, vertically interpolated to 200 hPa
Accessed in code	as umeridional_200hPa from the diag

umeridional_250hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 250 hPa
Accessed in code	as umeridional_250hPa from the diag

umeridional_500hPa (real) (nCells, Time)

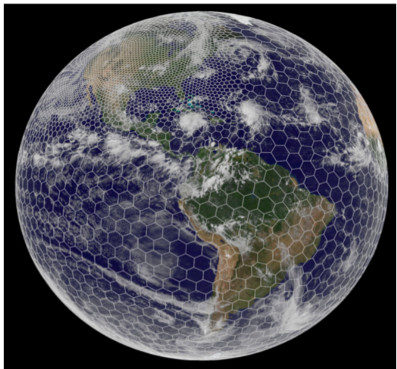
Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 500 hPa
Accessed in code	as umeridional_500hPa from the diag

umeridional_50hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 50 hPa
Accessed in code	as umeridional_50hPa from the diag

umeridional_6km (real) (nCells, Time)

Units	m s ⁻¹
Description	Meridional wind component at 6 km AGL
Accessed in code	as umeridional_6km from the diag



[MPAS Atmosphere](#)

Menu

umeridional_700hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 700 hPa
Accessed in code	as umeridional_700hPa from the diag

umeridional_850hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 850 hPa
Accessed in code	as umeridional_850hPa from the diag

umeridional_925hPa (real) (nCells, Time)

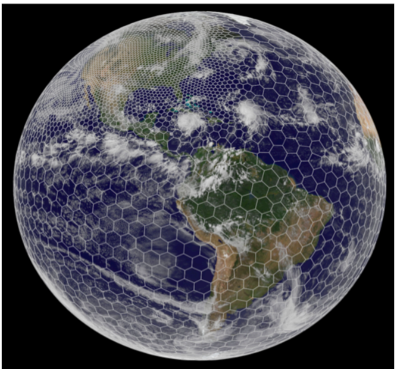
Units	m s ⁻¹
Description	Reconstructed meridional wind at cell centers, vertically interpolated to 925 hPa
Accessed in code	as umeridional_925hPa from the diag

umeridional_surface (real) (nCells, Time)

Units	m s ⁻¹
Description	Meridional wind component at midpoint of lowest model layer
Accessed in code	as umeridional_surface from the diag

updraft_helicity_max (real) (nCells, Time)

Units	m ² s ⁻²
Description	Maximum updraft helicity since last output



[MPAS Atmosphere](#)

Menu

Accessed in code	as updraft_helicity_max from the diag
---------------------	---

uReconstructMeridional (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	Meridional component of reconstructed horizontal velocity at cell centers
Accessed in code	as uReconstructMeridional from the diag



uReconstructX (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	Cartesian x-component of reconstructed horizontal velocity at cell centers
Accessed in code	as uReconstructX from the diag

uReconstructY (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	Cartesian y-component of reconstructed horizontal velocity at cell centers
Accessed in code	as uReconstructY from the diag

uReconstructZ (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	Cartesian z-component of reconstructed horizontal velocity at cell centers
Accessed in code	as uReconstructZ from the diag



[MPAS Atmosphere](#)

Menu

uReconstructZonal (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	Zonal component of reconstructed horizontal velocity at cell centers
Accessed in code	as uReconstructZonal from the diag

ust (real) (nCells, Time)

Units	m s ⁻¹
Description	U* in similarity theory
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as ust from the diag_physics

ustm (real) (nCells, Time)

Units	m s ⁻¹
Description	U* in similarity theory without vconv
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as ustm from the diag_physics

uzonal_100hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 100 hPa
Accessed in code	as uzonal_100hPa from the diag

uzonal_1km (real) (nCells, Time)

Units	m s ⁻¹
Description	Zonal wind component at 1 km AGL



[MPAS Atmosphere](#)

Menu

Accessed in code	as uzonal_1km from the diag
---------------------	--------------------------------------

uzonal_200hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 200 hPa
Accessed in code	as uzonal_200hPa from the diag

uzonal_250hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 250 hPa
Accessed in code	as uzonal_250hPa from the diag

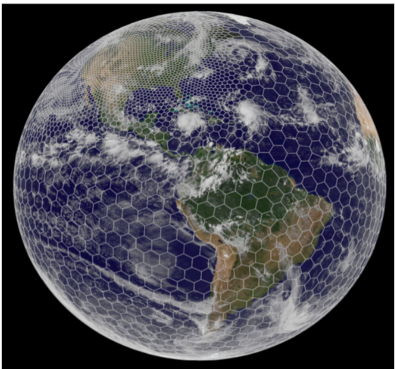
uzonal_500hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 500 hPa
Accessed in code	as uzonal_500hPa from the diag

uzonal_50hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 50 hPa
Accessed in code	as uzonal_50hPa from the diag

uzonal_6km (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	m s^{-1}
Description	Zonal wind component at 6 km AGL
Accessed in code	as uzonal_6km from the diag

uzonal_700hPa (real) (nCells, Time)

Units	m s^{-1}
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 700 hPa
Accessed in code	as uzonal_700hPa from the diag

uzonal_850hPa (real) (nCells, Time)

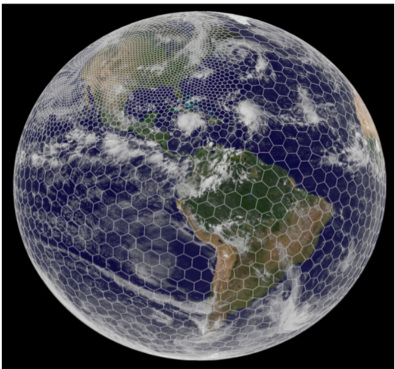
Units	m s^{-1}
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 850 hPa
Accessed in code	as uzonal_850hPa from the diag

uzonal_925hPa (real) (nCells, Time)

Units	m s^{-1}
Description	Reconstructed zonal wind at cell centers, vertically interpolated to 925 hPa
Accessed in code	as uzonal_925hPa from the diag

uzonal_surface (real) (nCells, Time)

Units	m s^{-1}
Description	Zonal wind component at midpoint of lowest model layer



[MPAS Atmosphere](#)

Menu

Accessed in code	as uzonal_surface from the diag
---------------------	---------------------------------------

v (real) (nVertLevels, nEdges, Time)

Units	m s ⁻¹
Description	Horizontal tangential velocity at edges
Accessed in code	as v from the diag

v10 (real) (nCells, Time)

Units	m s ⁻¹
Description	10-meter meridional wind
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as v10 from the diag_physics

v_init (real) (nVertLevels)

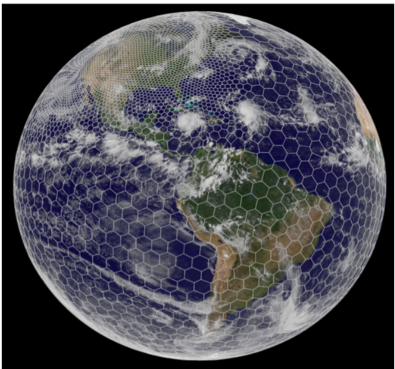
Units	m s ⁻¹
Description	v reference profile
Accessed in code	as v_init from the mesh

v_pv (real) (nCells, Time)

Units	m s ⁻¹
Description	Meridional wind on dynamic tropopause
Accessed in code	as v_pv from the diag

var2d (real) (nCells)

Units	m
Description	standard deviation of subgrid- scale orography
Accessed in code	as var2d from the sfc_input



[MPAS Atmosphere](#)

Menu

vegfra (real) (nCells, Time)

Units	unitless
Description	vegetation fraction
Accessed in code	as vegfra from the sfc_input

verticesOnCell (integer) (maxEdges, nCells)

Units	•
Description	IDs of vertices (corner points) of a cell
Accessed in code	as verticesOnCell from the mesh

verticesOnEdge (integer) (TWO, nEdges)

Units	•
Description	IDs of the two vertex endpoints of an edge
Accessed in code	as verticesOnEdge from the mesh

vort_pv (real) (nCells, Time)

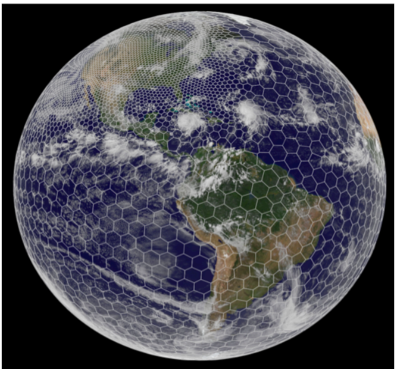
Units	s ⁻¹
Description	Relative vertical vorticity on dynamic tropopause
Accessed in code	as vort_pv from the diag

vorticity (real) (nVertLevels, nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity at vertices
Accessed in code	as vorticity from the diag

vorticity_100hPa (real) (nVertices, Time)

Units	s ⁻¹
-------	-----------------



[MPAS Atmosphere](#)

Menu

Description	Relative vorticity vertically interpolated to 100 hPa
Accessed in code	as vorticity_100hPa from the diag

vorticity_200hPa (real) (nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 200 hPa
Accessed in code	as vorticity_200hPa from the diag

vorticity_250hPa (real) (nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 250 hPa
Accessed in code	as vorticity_250hPa from the diag

vorticity_500hPa (real) (nVertices, Time)

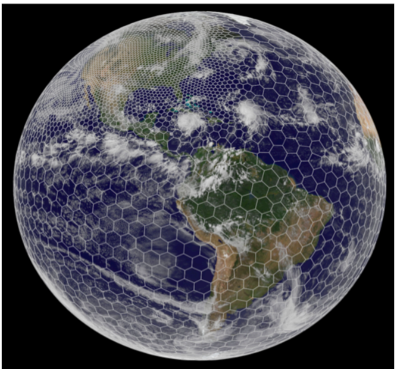
Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 500 hPa
Accessed in code	as vorticity_500hPa from the diag

vorticity_50hPa (real) (nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 50 hPa
Accessed in code	as vorticity_50hPa from the diag

vorticity_700hPa (real) (nVertices, Time)

Units	s ⁻¹
-------	-----------------



[MPAS Atmosphere](#)

Menu

Description	Relative vorticity vertically interpolated to 700 hPa
Accessed in code	as vorticity_700hPa from the diag

vorticity_850hPa (real) (nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 850 hPa
Accessed in code	as vorticity_850hPa from the diag

vorticity_925hPa (real) (nVertices, Time)

Units	s ⁻¹
Description	Relative vorticity vertically interpolated to 925 hPa
Accessed in code	as vorticity_925hPa from the diag

w (real) (nVertLevelsP1, nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity at vertical cell faces
Accessed in code	as w from the state

w0avg (real) (nVertLevels, nCells, Time)

Units	m s ⁻¹
Description	time running-averaged vertical velocity
Allocated by	cu_kain_fritsch_in
Accessed in code	as w0avg from the diag_physics

w_100hPa (real) (nCells, Time)

Units	m s ⁻¹
-------	-------------------



[MPAS Atmosphere](#)

Menu

Description	Vertical velocity vertically interpolated to 100 hPa
Accessed in code	as w_100hPa from the diag

w_200hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 200 hPa
Accessed in code	as w_200hPa from the diag

w_250hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 250 hPa
Accessed in code	as w_250hPa from the diag

w_500hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 500 hPa
Accessed in code	as w_500hPa from the diag

w_50hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 50 hPa



[MPAS Atmosphere](#)

Menu

Accessed in code	as w_50hPa from the diag
------------------	--------------------------

w_700hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 700 hPa
Accessed in code	as w_700hPa from the diag

w_850hPa (real) (nCells, Time)

Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 850 hPa
Accessed in code	as w_850hPa from the diag

w_925hPa (real) (nCells, Time)

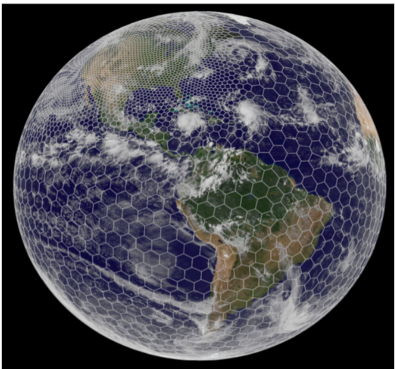
Units	m s ⁻¹
Description	Vertical velocity vertically interpolated to 925 hPa
Accessed in code	as w_925hPa from the diag

w_velocity_max (real) (nCells, Time)

Units	m s ⁻¹
Description	Maximum column w velocity since last output
Accessed in code	as w_velocity_max from the diag

weightsOnEdge (real) (maxEdges2, nEdges)

Units	unitless
-------	----------



[MPAS Atmosphere](#)

Menu

Description	Weights used in reconstruction of tangential velocity for an edge
Accessed in code	as weightsOnEdge from the mesh

wind_speed_level1_max (real) (nCells, Time)

Units	m s ⁻¹
Description	Maximum wind speed in lowest model level since last output
Accessed in code	as wind_speed_level1_max from the diag



wspd (real) (nCells, Time)

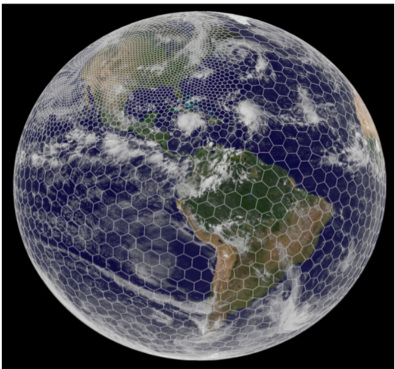
Units	m s ⁻¹
Description	wind speed at lowest model level
Allocated by	bl_mynn_in, bl_ysu_in
Accessed in code	as wspd from the diag_physics

wstar (real) (nCells, Time)

Units	m s ⁻¹
Description	mixed velocity scale from PBL scheme
Allocated by	bl_ysu_in
Accessed in code	as wstar from the diag_physics

wwAvg (real) (nVertLevelsP1, nCells, Time)

Units	kg m ⁻² s ⁻¹
Description	time-averaged rho*omega/zz used in scalar transport
Accessed in code	as wwAvg from the diag



[MPAS Atmosphere](#)

Menu

wwAvg_split (real) (nVertLevelsP1, nCells, Time)

Units	kg m ⁻² s ⁻¹
Description	time-averaged rho*omega/zz used in scalar transport
Accessed in code	as wwAvg_split from the diag

xCell (real) (nCells)

Units	m
Description	Cartesian x-coordinate of cells
Accessed in code	as xCell from the mesh

xEdge (real) (nEdges)

Units	m
Description	Cartesian x-coordinate of edges
Accessed in code	as xEdge from the mesh

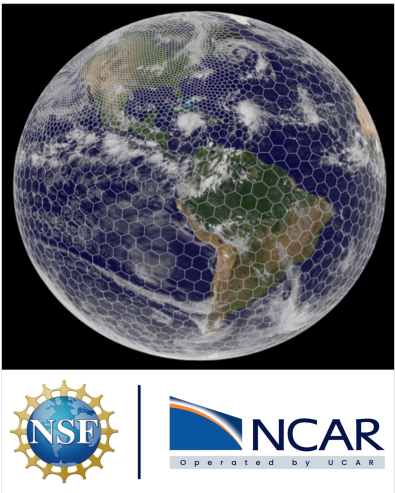
xice (real) (nCells, Time)

Units	unitless
Description	fractional area coverage of sea-ice
Accessed in code	as xice from the sfc_input

xicem (real) (nCells, Time)

Units	unitless
Description	sea-ice flag from previous time-step
Accessed in code	as xicem from the diag_physics

xland (real) (nCells, Time)



[MPAS Atmosphere](#)

Menu

Units	unitless
Description	land-ocean mask (1=land including sea-ice ; 2=ocean)
Accessed in code	as xland from the sfc_input

xmb_shallow (real) (nCells, Time)

Units	kg m ⁻² s ⁻¹
Description	cloud base mass flux for shallow convection
Allocated by	cu_grell_freitas_in
Accessed in code	as xmb_shallow from the diag_physics

xmb_total (real) (nCells, Time)

Units	kg m ⁻² s ⁻¹
Description	cloud-base mass flux
Allocated by	cu_grell_freitas_in
Accessed in code	as xmb_total from the diag_physics

xtime (text) (Time)

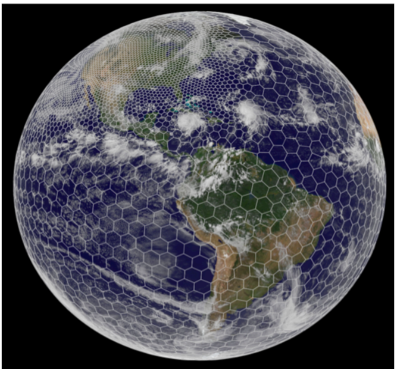
Units	YYYY-MM-DD_hh:mm:ss
Description	Model valid time
Accessed in code	as xtime from the state

xVertex (real) (nVertices)

Units	m
Description	Cartesian x-coordinate of vertices
Accessed in code	as xVertex from the mesh

yCell (real) (nCells)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Cartesian y-coordinate of cells
Accessed in code	as yCell from the mesh

yEdge (real) (nEdges)

Units	m
Description	Cartesian y-coordinate of edges
Accessed in code	as yEdge from the mesh

yVertex (real) (nVertices)

Units	m
Description	Cartesian y-coordinate of vertices
Accessed in code	as yVertex from the mesh

z0 (real) (nCells, Time)

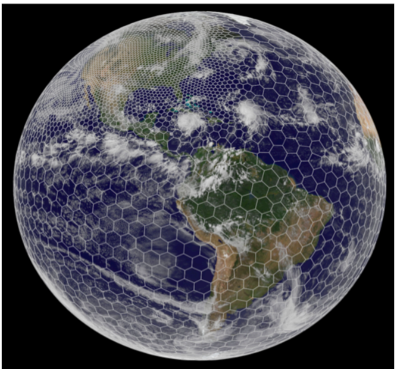
Units	m
Description	roughness height
Accessed in code	as z0 from the diag_physics

z_iso_levels (real) (nIsoLevelsZ)

Units	Pa
Description	Levels for vertical interpolation of height to isobaric surfaces
Accessed in code	as z_iso_levels from the diag

z_isobaric (real) (nIsoLevelsZ, nCells, Time)

Units	m
-------	---



[MPAS Atmosphere](#)

Menu

Description	Height interpolated to isobaric surfaces defined in z_iso_levels
Accessed in code	as z_isobaric from the diag

zb (real) (nVertLevelsP1, TWO, nEdges)

Units	unitless
Description	Coefficients for contribution from u to omega diagnosis, edge-oriented
Accessed in code	as zb from the mesh

zb3 (real) (nVertLevelsP1, TWO, nEdges)

Units	unitless
Description	Coefficients for 3rd-order correction to contribution from u to omega diagnosis, edge-oriented
Accessed in code	as zb3 from the mesh

zb3_cell (real) (nVertLevelsP1, maxEdges, nCells)

Units	unitless
Description	Coefficients for 3rd-order correction to contribution from u to omega diagnosis, cell-oriented



[MPAS Atmosphere](#)

Menu

Accessed in code	as zb3_cell from the mesh
------------------	---------------------------

zb_cell (real) (nVertLevelsP1, maxEdges, nCells)

Units	unitless
Description	Coefficients for contribution from u to omega diagnosis, cell-oriented
Accessed in code	as zb_cell from the mesh

zCell (real) (nCells)

Units	m
Description	Cartesian z-coordinate of cells
Accessed in code	as zCell from the mesh

zEdge (real) (nEdges)

Units	m
Description	Cartesian z-coordinate of edges
Accessed in code	as zEdge from the mesh

zgrid (real) (nVertLevelsP1, nCells)

Units	m MSL
Description	Geometric height of layer interfaces
Accessed in code	as zgrid from the mesh

znt (real) (nCells, Time)

Units	m
Description	roughness length



[MPAS Atmosphere](#)

Menu

Accessed in code	as znt from the diag_physics
------------------	------------------------------

zol (real) (nCells, Time)

Units	unitless
Description	z/L height over Monin-Obukhov length
Allocated by	bl_mynn_in, bl_yсу_in
Accessed in code	as zol from the diag_physics

zs (real) (nCells, Time)

Units	m
Description	depth of centers of soil layers
Accessed in code	as zs from the diag_physics

zVertex (real) (nVertices)

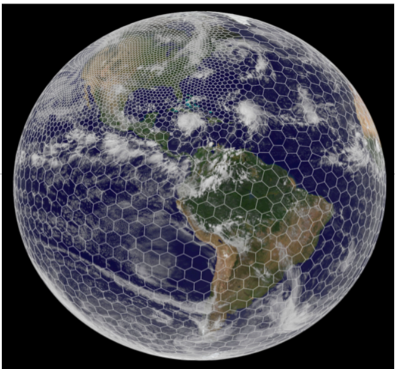
Units	m
Description	Cartesian z-coordinate of vertices
Accessed in code	as zVertex from the mesh

zxu (real) (nVertLevels, nEdges)

Units	unitless
Description	dz/dx on horizontal coordinate surfaces at u levels
Accessed in code	as zxu from the mesh

zz (real) (nVertLevels, nCells)

Units	unitless
Description	d(zeta)/dz, vertical metric term
Accessed in code	as zz from the mesh



[MPAS Atmosphere](#)

Menu

[TYPO3 Theme 4.9.0](#)

© Copyright 2024, NSF NCAR/MMM